
Forecasting Inflation in a Data-Rich Environment: The Benefits of Machine Learning Methods

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Abstract

This study aims to compare the performance of classical time series models with Machine Learning (ML) models and identify key predictors for forecasting US CPI inflation in a data-rich environment. The empirical analysis leads to the following key findings. First, ML methods, including Random Forest, XGBoost, Ridge regression, and adaptive LASSO, applied to a large dataset, achieve higher predictive accuracy than traditional time-series benchmarks across medium- to long-term forecasting horizons, particularly during high-volatility inflation periods. Among the competing models, Random Forest demonstrates superior prediction performance and maintains stable variable selection across horizons. Second, standalone tree-based models perform better than their hybrid specifications, suggesting that feature selection before model estimation does not necessarily enhance the model accuracy. Lastly, apart from inflation lags, which are the most influential and reliable predictive factors, other variables relevant to the housing market, the labor market, interest and exchange rates, and money and credit are also among the most important predictors for forecasting US CPI inflation. Altogether, these results contribute to macroeconomic time series forecasting and highlight the benefits of ML methods, together with big data, in achieving higher prediction accuracy.

Keywords: Inflation Forecasting, Machine Learning, Big Data, Random Forest, Variable Importance Analysis.

1 Introduction

Inflation is a persistent increase in the general price level of goods and services in an economy over time. It is considered one of the most important key economic indicators. The future inflation rate can heavily affect households, economic agents, authorities, and policymakers in making optimal economic decisions and investment strategies over time. High inflation rates weaken economic growth, reduce real wages and savings, and raise production costs. In contrast, low rates indicate lower demand in the economy, as indicated by lower spending and consumption and longer investment project durations. Central banks usually rely on inflation forecasts to implement effective economic policies and to anchor the expected inflation rate, thereby stabilizing the economy. Therefore, producing accurate inflation forecasts plays a crucial role in economic decision-making. Inflation forecasting is a difficult task. This mainly stems from the presence of high correlation, nonlinearity, and high dimensionality among the macroeconomic and financial variables, as well as potential structural breaks in the data, which challenge traditional forecasting techniques and the formulation of a forecasting equation. ([Aras and Lisboa \(2022\)](#) and [Doojav and Narmandakh \(2025\)](#)). However, the combination of recent advancements in ML models and the availability of big datasets appears to be a potential alternative solution to these issues.

Most of the previous inflation forecasting literature has widely agreed that multivariate time-series models outperform univariate models. However, the use and benefits of ML models, combined with large datasets, have been discussed only sparsely in these studies on macroeconomic forecasting. In particular, One of the seminal studies is [Atkeson et al. \(2001\)](#) evaluates the usefulness of the Phillips curve-based models for inflation forecasting based on real economic activity, but eventually, the authors argued that the search for other augmented Phillips-curve models should

be abandoned due to their lower forecast accuracies compared to naive models and the rising complexity of inflation in the United States over the past 15 years according to [Stock and Watson \(2007\)](#). On the other hand, the study in [Stock and Watson \(1999\)](#) finds that the Phillips curve, with different specifications, yields more reliable and stable inflation forecasts than univariate forecasting models. Still, their work is limited by relying solely on linear models and assuming a linear relationship between inflation and the candidate variables. No forecasting horizon longer than 1 year is used to evaluate the models. As a result, existing research on inflation forecasting is inadequate for modeling the complex relationship between covariates and inflation, leading to inaccurate predictions that can undermine the effectiveness of the central bank's policy.

In recent years, due to advancements in computing power and the availability of larger datasets, ML models have gained increasing attention, and economists and researchers have considered them potential alternatives to classical time-series models to enhance inflation forecasting accuracy. Several studies have been conducted to verify this finding. One of them is [Nakamura \(2005\)](#) showed that on average, the neural networks outperform univariate autoregressive models for short horizons when forecasting U.S. inflation. Another finding of [Garcia et al. \(2017\)](#) revealed that high-dimensional econometric models have excellent performance in real-time inflation forecasting within a data-rich environment in Brazil, confirming the usefulness of big data in macroeconomic forecasting. However, [Medeiros et al. \(2021\)](#) raised a question about whether similar findings of this study can be replicated for the case of the U.S. economy. [Medeiros et al. \(2021\)](#) with a comprehensive analysis found that ML models incorporating a large number of covariates systematically outperform the univariate benchmark models in terms of inflation forecasting accuracy, and the results of this paper also confirm the superiority of random forest models to predict U.S CPI inflation rates. Numerous studies show a preference for ML techniques in data-rich environments. For instance, the empirical work of [Doojav and Narmandakh \(2025\)](#) indicates that ML approaches such as XGBoost, Random Forest, and Ridge Regression, when applied to large datasets, produce more accurate inflation forecasts than AR and factor models. This paper also emphasizes the strength of ML models in handling nonlinearities, multicollinearity, and dimensionality. Another result from [Huang et al. \(2025\)](#) showed that ML methods based on a large number of macroeconomic and financial predictors are superior for inflation forecasting in China compared to conventional time series, especially during periods of high inflation volatility. This paper also highlights that penalized linear regression models, including ridge and elastic net, are worth considering given their outstanding inflation-forecast performance. A narrow replication of [Naghi et al. \(2024\)](#) confirms the results of [Medeiros et al. \(2021\)](#) on the dominance of Random Forest for US inflation forecasting; however, this finding is not universal when considering extended sets of models and US data, where other competing tree-based models with better forecast performance surpass RF.

Given the lack of research regarding inflation forecasting, this study aims to compare and evaluate the inflation forecasting performance of benchmark models, including Autoregressive model, Random Walk, AORW, Generalized Autoregressive Score model, and Machine Learning models such as LASSO and Ridge Regression and adaptive Lasso, Elastic Net, Random Forest, XGBoost, and hybrid ML models across different forecast horizons within a data-rich environment using the updated FRED-MD monthly macroeconomic data for the U.S economy, as compiled by [McCracken and Ng \(2016\)](#), and examine the importance of ML approaches in forecasting U.S. inflation. To do so, this study addresses these gaps by first conducting a model comparison between traditional

time series models and ML forecasting methods across different forecasting horizons in the full out-of-sample period and two subsamples of more- and less-volatile inflation periods. Second, this analysis examines ML hybrid specifications to assess the usefulness of feature-selection techniques applied before model estimation in enhancing the quality of inflation forecasts. Finally, this thesis aims to identify the features with the greatest explanatory power for the US inflation forecasting exercise, based on the best-performing ML models. This study will contribute to the inflation-forecasting literature, which can be summarized as follows: Firstly, it provides a comprehensive comparison and assessment of ML models for US CPI inflation forecasting across different horizons, using both full-sample and subsample analyses. Secondly, this work can better identify the most informative predictors, potentially serving as key drivers of U.S. inflation dynamics, and determine the contribution of each selected feature to the prediction problem. The final contribution is to test whether feature selection before model estimation improves the inflation-forecasting performance of ML models. These findings can further enhance inflation forecasting accuracy and, therefore, help central banks achieve optimal monetary policies that stabilize the inflation rate. However, the limitation of this study is that it cannot completely guarantee that my findings can be generalized to other countries due to differences in each country's economy, inflation dynamics, and predictor sets, which lead to distinct volatility patterns in inflation rates; hence, achieving desired accuracy in inflation forecasting for other countries is no longer possible.

The rest of this study is constructed as follows. Sections II and III describe the methodology, which includes an inflation-forecasting equation, traditional econometric methods, and ML models. Section IV introduces the data and hyperparameter tuning. Section V provides the empirical results in comparing and evaluating the forecasts of competing models. Section VI is the conclusion. The appendix provides additional stationary transformation tables in data processing steps.

2 Methodology

This section describes the methods and models used in this study to forecast future U.S. CPI inflation. Our main goal is to forecast the inflation rate π_{t+h} at month $t + h$ using the information set available at month t . A direct multi-step-ahead forecasting framework is considered for future inflation because this approach is more robust to model misspecification, in contrast to the iterated or recursive forecasting method, which is constructed iteratively using one-step-ahead forecasting model's estimators (Marcellino et al. (2006)). Another reason for choosing the direct forecast approach is to avoid estimating a model for the time evolution of the predictors x_t and to circumvent specification errors in recursive settings, as discussed in Garcia et al. (2017) and Huang et al. (2025). We consider the following inflation forecasting model.

$$\pi_{t+h} = F_h(\mathbf{X}_t) + u_{t+h}, \quad h = 1, \dots, H, \quad t = 1, \dots, T.$$

Where π_{t+h} is the inflation at month $t + h$, where T denotes the full sample size and h is the forecasting horizon, where the horizons of 1-, 3-, 6-, 9- and 12-month-ahead forecasts are considered in this paper. $\mathbf{X}_t = (x_{1t}, \dots, x_{nt})'$ is an $(n \times 1)$ vector of predictor variables, containing lags of inflation π_t , and a large set of macroeconomic explanatory variables up to time t . $F_h(\cdot)$ is a function mapping the predictors $\mathbf{X}_t$ to the future inflation π_{t+h} , u_{t+h} is a zero-mean random error. The target function $F_h(\mathbf{X}_t)$ can be a time series model, an ML model, or a hybrid ML model, and there is a different functional mapping for each forecasting horizon (Medeiros et al. (2021) and Doojav and Narmandakh (2025)).

The direct multi-step-ahead forecasting approach is based on the rolling-window schemes of fixed-length. This rolling window procedure not only reduces the impacts of potential structural breaks and outliers, but it also helps capture the model specification uncertainty, model instability, estimation uncertainty, and the uncertainty of future events during the model estimation process using the data up to date t (Medeiros et al. (2021) and Huang et al. (2025)). This is indicated through the following direct forecasting equation form

$$\hat{\pi}_{t+h|t} = \hat{F}_{h, t-R_{h+1}:t}(\mathbf{X}_t).$$

Where $\hat{F}_{h, t-R_{h+1}:t}$ is the estimated target function using the data that spans from time $t - R_h + 1$ until time t , and R_h is the size of the window, and $R_h = N - h$, which is computed once for each forecasting horizon by subtracting the horizon h from the number of in-sample observations N before the out-of-sample date in January 1990. Forecasting inflation is constructed using different models $F_h(\mathbf{X}_t)$, including benchmark models (AR, Random Walk, AORW, GAS) and supervised Machine Learning models consist of the penalized linear regression models (Ridge Regression, LASSO, Elastic Net, adaptive LASSO), tree-based models (Random Forest (RF), and extreme gradient boosting (XGBoost)), and additional hybrid ML models (adaLASSO-RF, adaLASSO-XGBoost) to verify the importance of preselected feature in enhancing model forecasting ability. For the shrinkage, the set of predictors is standardized before model estimation. In contrast, feature scaling does not apply to plain tree-based models, which are considered a common practice according to Araujo and Gaglianone (2023). In the case of hybrid ML models, scaled predictors are used only in the first step of adaptive LASSO feature preselection, which returns the indices of relevant variables. RF and XGBoost are applied to these selected unscaled features. These models are described in the section below.

3 Models

3.1 Benchmark Models

In this subsection, the set of benchmark models selected for inflation forecasting is considered from the work of Medeiros et al. (2021) and Huang et al. (2025). We denote π_t as the month-on-month inflation at time t .

3.1.1 Random Walk Model (RW)

The first benchmark model is the Random Walk model (RW). For the forecast horizons $h = 1, 3, 6, 9, 12$, the inflation forecast of RW for period $t + h$ made at time t is $\hat{\pi}_{t+h|t} = \pi_t$, which is the last observed value of inflation.

3.1.2 Autoregressive Model (AR(p))

The second benchmark model is the autoregressive model of order p , AR(p). The lag order p is determined by the Bayesian Information Criterion (BIC) for each window with the defined search space is $p \in \{1, 2, 3, 4\}$, as discussed by Doojav and Narmandakh (2025); this is different from the case of Medeiros et al. (2021) and Huang et al. (2025), who fix the inflation lag length at four. The parameters are estimated by OLS. The forecast equation is as follows:

$$\hat{\pi}_{t+h|t} = \hat{\beta}_{0,h} + \hat{\beta}_{1,h} \pi_t + \cdots + \hat{\beta}_{p,h} \pi_{t-p+1}.$$

Where $\hat{\pi}_{t+h|t}$ represents the h -month-ahead forecasted inflation given the information available up to date t . The parameters $\hat{\beta}_{i,h}$ for $i = 0, 1, \dots, p$ are OLS estimates and $\pi_t, \dots, \pi_{t-p+1}$ are the lagged values of inflation. This time-series model assumes a linear relationship between the target variable as the future inflation rate, and its p -lagged values.

3.1.3 AORW Model

The third benchmark is the AORW forecasting model, which is essentially a variant of the Random Walk forecast considered by [Atkeson et al. \(2001\)](#). Following the findings of [Stock and Watson \(2007\)](#), we extend the AORW forecast to other horizons without adjustments, in particular by averaging inflation over the last 12 months. Therefore, the AORW forecast is defined as follows:

$$\pi_{t+h|t}^h = \pi_t^{12} = \frac{1}{12} (\pi_t + \pi_{t-1} + \dots + \pi_{t-11})$$

This benchmark univariate model is also employed in the studies of forecasting inflation in China, as in [Huang et al. \(2025\)](#), and in the case study of Brazil [Araujo and Gaglianone \(2023\)](#).

3.1.4 Generalized Autoregressive Score Model (GAS(1,1))

The final benchmark is the GAS(1,1) model, an observation-driven time series model introduced by [Creal et al. \(2013\)](#) and [Harvey \(2013\)](#). The main idea behind this model is to update the parameter over time. This is achieved by the scaled score of the likelihood with a chosen scaling matrix, where the score defines the steepest ascent direction for improving the model's local fit. Applying this to the US CPI inflation forecasting framework, we formulate the model specification as follows. Let π_t be the dependent variable of interest at time t ; the time-varying parameter here is set as the mean f_t of inflation; the variance is assumed to be constant σ^2 , and $\theta = (\omega, A, B, \sigma^2)$ is a vector of static parameters. The reason behind this setup is that this study concentrates on predicting the average inflation level in % instead of inflation volatility; therefore, the conditional mean f_t is chosen to move over time, and the variance is fixed, and the lag orders are fixed at $p = 1$ and $q = 1$ to have a parsimonious model. We have the distribution of the target variable π_t given the available information $\mathcal{F}_{t-1}$ is Gaussian with a conditional mean and constant variance.

$$\pi_t | \mathcal{F}_{t-1} \sim \mathcal{N}(f_t, \sigma^2), \quad \pi_t = \log(P_t) - \log(P_{t-1}), \quad t = 1, \dots, T$$

The observation density and the corresponding log-likelihood function are therefore given by

$$\pi_t \sim p(\pi_t | f_t, \mathcal{F}_{t-1}, \theta)$$

$$\ln p(\pi_t | f_t, \mathcal{F}_{t-1}, \theta) = \ell_t = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(\pi_t - f_t)^2}{2\sigma^2}$$

The updating equation for the GAS(1,1) model is

$$f_{t+1} = \omega + A s_t + B f_t$$

where ω is a constant, A and B are coefficients. The recursion is initialized at the in-sample mean of each window, $f_1 = \bar{\pi}$. Then the time-varying parameter f_t is updated to the next period $t + 1$

with the scaled score, where the score is defined as the derivative of the log-likelihood with respect to the mean f_t

$$\nabla_t = \frac{\partial \ell_t}{\partial f_t} = \frac{\pi_t - f_t}{\sigma^2}$$

The scaling matrix S_t is set equal to the inverse of the information matrix.

$$S_t = I_{t|t-1}^{-1} = \sigma^2, \quad I_{t|t-1} = \mathbf{E}_{t-1}[\nabla_t^2] = \frac{1}{\sigma^2}$$

The scaled score is the driving mechanism of the GAS(1,1) model; in this case, it is defined as the forecast error.

$$s_t = S_t \cdot \nabla_t = \sigma^2 \cdot \frac{\pi_t - f_t}{\sigma^2} = \pi_t - f_t$$

It is now clear to see that the choice of inverse information matrix of $S_t = \sigma^2$ simplifies the raw score ∇_t to the simple forecast error term $\pi_t - f_t$, avoiding the chance of dividing by a tiny value of variance σ^2 , which can potentially lead to numerical instability. The maximum likelihood approach is used to estimate the static parameter vector.

$$\hat{\theta} = (\hat{\omega}, \hat{A}, \hat{B}, \hat{\sigma}^2) = \arg \max_{\theta} \sum_{t=1}^T \ell_t$$

Replacing s_t with the obtained forecast error, we have the recursion for the GAS(1,1) model:

$$f_{t+1} = \omega + A(\pi_t - f_t) + Bf_t = \omega + A\pi_t + (B - A)f_t$$

This can be interpreted as an error-correction average inflation model, with a current forecast of average inflation f_t for each month t . This current forecast is then corrected and updated via the forecast error, and this continues until the end of the sample.

The update equation of GAS(1,1) model is as follows:

$$f_{t+1} = \omega + As_t + Bf_t, \quad s_t = \pi_t - f_t$$

Since the score $s_t = \pi_t - f_t$ is the forecast error, the update equation only remains valid if the observed inflation π_t at month t is available. However, this is not the case when running this equation beyond the in-sample window. In that case, the future scaled score s_{t+j} is replaced with the conditional expectation

$$E_t[s_{t+j}] = E_t[\pi_{t+j} - f_{t+j}] = 0 \quad \text{for all } j \geq 1, \quad \text{where } \pi_{t+j} | \mathcal{F}_{t+j-1} \sim N(f_{t+j}, \sigma^2),$$

Therefore, the score term $A \cdot s_t$ or forecast error disappears for subsequent forecasts. The recursion for the GAS(1,1) model therefore reduces to only $\omega + B \cdot f_t$. The one-month-ahead inflation forecast update equation using the observed inflation rate is as follows:

$$\hat{f}_{t+1|t} = \omega + A(\pi_t - f_t) + Bf_t$$

The scaled score term is removed because no realized inflation rates are available beyond month t for subsequent inflation forecasts.

$$\hat{f}_{t+j|t} = \omega + B\hat{f}_{t+j-1|t}, \quad j = 2, \dots, h$$

Unfolding this recursion yields the final closed-form expression for the h -step-ahead inflation forecasting framework.

$$\hat{f}_{t+h|t} = \omega \frac{1 - B^{h-1}}{1 - B} + B^{h-1} \hat{f}_{t+1|t} = \hat{\pi}_{t+h|t},$$

Where $B \neq 1$. Moreover, if $|B| < 1$, then the inflation forecasts of GAS(1,1) converge to unconditional mean $\frac{\omega}{(1-B)}$ as the horizon h increases. Thus, future inflation forecasts of the GAS model are obtained when conditional on the information available through month t .

3.2 Penalized Linear Regression Models

Penalized regression is also known as shrinkage or regularization techniques. These methods introduce a linear regression model with penalties for having too many predictors, which can lead to overfitting. When there are many correlated predictors, or when the number of explanatory variables N is much higher than the number of observations T , overfitting can be considered a serious problem. In this case, although the in-sample model predictive performance is accurate, its out-of-sample forecasts are inaccurate. To deal with this issue, regularization, which proposes penalty terms, can eliminate the less contributive variables by reducing the coefficient values of the less relevant predictors close to or equal to zero.

Given the direct forecasting equation $\pi_{t+h} = F_h(\mathbf{X}_t) + u_{t+h}$, we have $F_h(\cdot)$ is in the format of a linear regression model where

$$F_h(\mathbf{X}_t) = \beta'_h \mathbf{X}_t = \beta_{0,h} + \sum_{i=1}^N \beta_{i,h} x_{it}$$

These models minimize the following general penalized regression defined as follows:

$$\hat{\beta}_h = \arg \min_{\beta_h} \left[\sum_{t=1}^{T-h} (\pi_{t+h} - \beta'_h \mathbf{X}_t)^2 + \sum_{i=1}^N p(\beta_{h,i}; \lambda, \omega_i) \right]$$

Where T is the total number of observations and N is the number of predictors, $p(\beta_{h,i}; \lambda, \omega_i)$ corresponds to a penalty function that depends on the shrinkage parameter λ and on a weight $\omega_i > 0$. The value of the hyperparameter λ determines the magnitude of the penalty term and the bias-variance tradeoff. A higher value of λ means a greater shrinkage of the regression coefficients β_h , resulting in more bias and less variance in the forecasting model. For the shrinkage-based models, the set of predictors is standardized before estimation.

We consider different choices for the penalty functions $p(\beta_{h,i}; \lambda, \omega_i)$. In this study, we consider four main shrinkage models for forecasting US CPI inflation, including the least absolute shrinkage and selection operator (LASSO), adaptive LASSO (adaLASSO), Ridge Regression (RR), and Elastic Net (EINet) models.

3.2.1 Ridge Regression (RR)

Ridge Regression is proposed by [Hoerl and Kennard \(1970\)](#). The objective of RR is to deal with highly correlated regressors by reducing the variance of the estimator in exchange for a small bias. RR shrinks the regression coefficients of variables with a tiny contribution to the outcome to nearly

zero. This is achieved by penalizing the regression model with an L2-norm penalty term, which is defined below:

$$\sum_{i=1}^N p(\beta_{h,i}, \lambda, \omega_i) := \lambda \sum_{i=1}^N \beta_{h,i}^2$$

RR has the advantage of an analytic solution, which allows effortless computation. However, the coefficients obtained by ridge regression are very small but never exactly equal to zero, regardless of any value of λ . This means that Ridge is not appropriate for feature selection. An interesting fact is that RR is a special case of the Elastic Net if $\alpha = 1$ in the penalty function (1).

3.2.2 The Least Absolute Shrinkage and Selection Operator (LASSO)

The LASSO regression model is an alternative method proposed by Tibshirani (1996). The core idea is to shrink the coefficients of irrelevant variables exactly to zero, thereby overcoming the problem of ridge regression by imposing the penalty function called the L1 norm on the regression coefficients $\beta_{h,i}$, defined as follows:

$$\sum_{i=1}^N p(\beta_{h,i}; \lambda, \omega_i) := \lambda \sum_{i=1}^N |\beta_{h,i}|$$

In this case, LASSO can also perform as a variable selection technique. LASSO can handle the case where there are more variables N than observations T by identifying the relevant subset of features. Nonetheless, LASSO also has limitations, according to Zou and Hastie (2005); it may be an inappropriate method for feature selection in some cases. One of them is the grouping effect, which happens when the regression coefficients of a group of highly correlated variables tend to be equal. Another case is when $T > N$ when the number of observations is larger than the number of highly correlated predictors, RR tends to outperform LASSO. LASSO is also a special case of the Elastic Net, considering $\alpha = 0$ in the Elnet penalty equation (1).

3.2.3 adaptive LASSO (adaLASSO)

The adaptive LASSO regression model, proposed by Zou (2006), aims to reduce estimation bias, achieve variable selection consistency, and improve forecasting accuracy compared to the standard LASSO, which can be inconsistent in selecting variables under certain circumstances. AdaLASSO uses the same penalty term as LASSO, but includes an additional weighting parameter ω_i . The core idea behind the model is to use prior information to select variables more effectively. AdaLASSO is a two-step estimation procedure that first generates a different weight ω_i for each candidate variable $x_{i,t}$. These weights are then used in the second step of LASSO estimation as additional information. The penalty function is defined as follows:

$$\sum_{i=1}^N p(\beta_{h,i}; \lambda, \omega_i) := \lambda \sum_{i=1}^N \omega_i |\beta_{h,i}|$$

Where $\omega_i = |\tilde{\beta}_{h,i}|^{-1}$ represents the weights and $\tilde{\beta}_{h,i}$ is the respective LASSO coefficient estimated in the first step. Hence, adaLASSO can assign a higher penalty to zero coefficients and a lower penalty to nonzero ones, handle many more candidate variables N than observations T , and perform well even under non-Gaussian and heteroskedastic errors. When implementing this model,

we set $\omega_i = \frac{1}{|\hat{\beta}_{h,i}| + \frac{1}{\sqrt{T}}}$ to avoid zero denominator as the adaLASSO would set the coefficients of variables exactly to zero and adding the term $\frac{1}{\sqrt{T}}$ would prevent division by zero $\frac{1}{|\hat{\beta}_{h,i}|}$ (Medeiros and Mendes (2016), Araujo and Gaglianone (2023) and Huang et al. (2025)).

3.2.4 Elastic Net (EINet)

The Elastic Net model is a generalization that includes RR and LASSO as special cases, and was proposed by Zou and Hastie (2005). In particular, it is a regression model penalized by both the L1- and L2-norm penalty functions. This resulting combination leads to an effective shrinkage of coefficients towards zero (as in RR) and some coefficients being set to exactly zero (as in LASSO). The EINet penalty function is defined as follows:

$$\sum_{i=1}^N p(\beta_{h,i}; \lambda, \omega_i) := \alpha \lambda \sum_{i=1}^N \beta_{h,i}^2 + (1 - \alpha) \lambda \sum_{i=1}^N |\beta_{h,i}| \quad (1)$$

Where $\alpha \in [0, 1]$ is the hyperparameter to determine the relative weights of the two penalty terms, L1 and L2 norms. Elastic Net benefits from the useful properties of Ridge and LASSO simultaneously, namely, shrinkage and sparsity. L2-norm regularization allows Elastic Net to handle many highly correlated predictors, while L1-norm regularization takes advantage of sparsity for the algorithm (Huang et al. (2025) and Aras and Lisboa (2022)).

The above shrinkage models (Ridge Regression, LASSO, adaptive LASSO, and Elastic Net) cannot capture nonlinear relationships between a target variable and the predictors. Therefore, nonlinear ML models, Random Forest (RF), Extreme Gradient Boosting (XGBoost), and hybrid ML models are considered in this U.S. inflation forecasting analysis.

3.3 Tree-based ML Models

3.3.1 Random Forest (RF)

The Random Forest (RF), introduced by Breiman (2001), is an ensemble learning model that consists of decision trees used for both regression and classification problems. The main idea is to enhance predictive accuracy, alleviate the overfitting that regression trees often face, and reduce their high prediction variance by using bootstrap aggregation (bagging) of randomly constructed regression trees. In turn, a regression tree is a nonparametric model based on the recursive binary partitioning of the covariate space X . The model is usually displayed as a graph of a binary decision tree, with the root node as the base of the tree, parent or split nodes, and terminal nodes, also called leaves, representing different partitions of X (Araujo and Gaglianone (2023) and Garcia et al. (2017)). The procedure for the Random Forest model can be summarized as follows. Firstly, multiple decision trees are trained on bootstrapped samples of the data. Each regression tree uses a random subset of features at each node. As a result, the trees are as different as possible, since they learn different data patterns and use different subsets of predictors. The final output is obtained by taking the average of the individual tree predictions. According to Medeiros et al. (2021) and Huang et al. (2025), the forecasting framework for Random Forest prediction for inflation can be formalized as follows:

First, B distinct bootstrap samples are drawn from the original data. Then, a random subset of the feature vector $\mathbf{X}_t$ is selected for each sample $b = 1, 2, 3, \dots, B$. The decision tree uses one of the

chosen variables to split the feature space into two regions at $X_1 = s_1$ and further partitions the covariate space into more non-overlapping regions R_k , where $k = 1, 2, \dots, K$ and each region corresponds to a terminal node of the decision tree. We then repeat the above process until the terminal nodes, or leaves, are too small to split further. The block bootstrap is used in this study because of the time-series nature of the dataset, as noted in [Medeiros et al. \(2021\)](#). Given the dependent variable π_{t+h} , feature set X_t , and the number of terminal nodes K , for each terminal node R_k with $1 \leq k \leq K$, the splits are computed by minimizing the sum of squared errors of the following regression model:

$$\pi_{t+h} = \sum_{k=1}^K c_k I\{x_t \in R_k\}$$

$$I(x_t \in R_k) = \begin{cases} 1, & x_t \in R_k, \\ 0, & \text{otherwise.} \end{cases}$$

Where c_k is a constant that corresponds to the k th region, which is estimated as the sample average of the realizations of inflation falling into region R_k , i.e. $\hat{c}_k = \text{mean}(\pi_{t+h}^k)$ and $I(x_t \in R_k)$ is the indicator function. The final forecast is the average of the forecasts of each tree K_b applied to the original data:

$$\hat{\pi}_{t+h} = \frac{1}{B} \sum_{b=1}^B \left[\sum_{k=1}^{K_b} \hat{c}_k I\{x_t \in R_k\} \right]$$

Where $\hat{\pi}_{t+h}$ is the predicted h -step-ahead inflation. According to [Doojav and Narmandakh \(2025\)](#), the RF algorithm is able to balance predictive accuracy and interpretability. It also excels at capturing nonlinear interactions, and these are crucial for understanding the complex dynamics of inflation. Random forest is also considered a reliable feature selection technique since Random Forest evaluates and extracts a random subset of features at each split, as advocated by [Aras and Lisboa \(2022\)](#). However, RF also has some limitations due to intensive and time-consuming computation ([Dewi and Chen \(2019\)](#)).

3.3.2 Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) is a decision-tree-based ensemble algorithm that uses a gradient-boosting setup, as proposed in [Chen and Guestrin \(2016\)](#). Its core idea is to iteratively add a weak learner (a decision tree) at a time to minimize the loss function. In other words, XGBoost builds decision trees sequentially. In particular, each new tree is added to reduce the errors of the previous one, a process known as boosting. The XGBoost model can be mathematically represented as follows:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i)$$

where the XGBoost algorithm uses K additive functions to predict the output, $\hat{y}_i$ is the final predicted value, K is the number of trees, and $f_k(x_i)$ stands for the prediction of the k^{th} tree. XGBoost minimizes the following regularized objective function:

$$\mathcal{L}(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k)$$

Where $l(\hat{y}_i, y_i)$ is the differentiable loss function which measures the difference between the true value y_i and the predicted value $\hat{y}_i$, the second term $\Omega(f_k)$ is the regularization term that penalizes overly complex trees. XGBoost is considered in this analysis for the following reasons. One of them is to prevent overfitting and improve the model's generalization by adding penalty terms that control tree complexity. The second is that it uses parallel processing for faster training, given the large monthly U.S. macroeconomic dataset. Therefore, XGBoost offers superior predictive performance in both generalization and speed. However, a shortcoming of this algorithm is that XGBoost has many hyperparameters that highly affect its predictive performance, as noted in [Aras and Lisboa \(2022\)](#).

3.4 Hybrid ML Models

A hybrid inflation-forecasting method is used to investigate whether the predictive accuracy of a given ML model is primarily determined by variable selection. Inspired by the approach of [Medeiros et al. \(2021\)](#), we conduct the following steps:

1. Within each window, adaptive LASSO is employed to choose the most relevant predictors with non-zero coefficients. However, a safety condition is set to select the top two features with the largest absolute coefficient values when adaLASSO selects fewer than two variables.
2. Using the set of top variables according to adaLASSO as input in the tree-based models: RF and XGBoost.
3. Tuning and estimating these ML models to forecast the US CPI inflation rate.

In this way, we have two hybrid ML models, including adaLASSO-RF and adaLASSO-XGBoost. Although the set of hybrid forecasting approaches can be further expanded, I believe these two hybrid models are sufficient to test the impact of feature selection before model estimation on improving ML forecasting performance.

3.5 Statistical Test

3.5.1 Diebold-Mariano (DM) Test

The Diebold-Mariano (DM) test is a statistical tool proposed by [Diebold and Mariano \(2002\)](#) used to compare the accuracy of two competing forecasting models. In particular, this test is formulated using the loss differential procedure, which involves choosing an appropriate loss function for forecast errors. In this study, I apply a two-sided DM test to assess the relative predictive ability of the superior model against individual benchmarks for each forecasting horizon using the full out-of-sample data. This is conducted in the following steps

1. Define the forecast errors as $\hat{u}_{it} = \hat{\pi}_{it} - \pi_t$, $i = 1, 2$ and $t = 1, \dots, T$, where $i = 1$ represents the best-performing model (RF) and $i = 2$ each of the benchmark models (RW, AR, GAS, AORW).
2. The loss differential between the two forecasts is then constructed as $d_t = g(u_{1t}) - g(u_{2t})$, where $g(\cdot)$ is the loss function, and the square-error loss $g(u_{it}) = u_{it}^2$ and absolute-error loss $g(u_{it}) = |u_{it}|$ of the forecast errors are considered.

3. The null hypothesis of equal predictive accuracy versus the alternative hypothesis of different predictive accuracy is

$$H_0 : E(d_t) = 0 \quad \forall t \quad vs \quad H_1 : E(d_t) \neq 0$$

4. The DM test statistic under H_0 is defined as

$$DM = \frac{\bar{d}}{\hat{\sigma}_{\bar{d}}}$$

where $\bar{d} = \frac{1}{T} \sum_{t=1}^T d_t$ is the sample mean of the loss differential series and the standard errors $\hat{\sigma}_{\bar{d}} = \sqrt{\widehat{\text{Var}}(\bar{d})}$ is based on the HAC estimators with $h-1$ lags for the autocovariance (Medeiros et al. (2021)). I use significance levels of 1%, 5%, and 10%.

5. Small-sample adjustment with Student-t distribution with $T-1$ degrees of freedom according to Harvey et al. (1997) is implemented to handle the finite-sample issue.

4 Data Description and Hyperparameter Tuning

4.1 Data

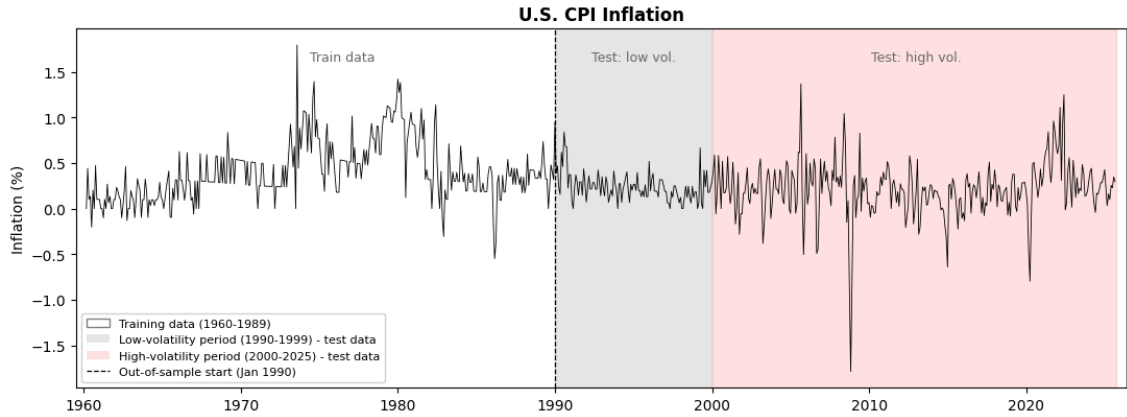


Figure 1: Time Series Plot for Monthly U.S. CPI Inflation over the Full Sample Period

Table 1: Descriptive statistics of the forecasted U.S. CPI inflation

	Mean	Sd	Max	Min
Low Volatility Period: 1990–1999	0.242	0.163	0.946	0.000
High Volatility Period: 2000–2025	0.211	0.302	1.367	-1.786

The data used in this study is extracted from the FRED-MD database, which is a large monthly macroeconomic dataset for the US, designed for empirical analysis in a data-rich environment, and it has been widely used in macroeconomic time series forecasting, as advocated by Medeiros et al. (2021) and Huang et al. (2025). The dataset is compiled and updated by McCracken and Ng (2016). I use the vintage data of May 2026. The sample is fixed to span from January 1960 to September 2025 (789 observations), since there are missing observations in CPI and 20 other

series in October 2025, so ending at September 2025 gives me a complete dataset at the cost of losing a few months, while the sample starts following [Medeiros et al. \(2021\)](#). I then conduct the following data preprocessing steps. Firstly, only variables with complete observations over the sample period are retained (119 variables). Second, all variables are transformed to ensure stationarity using the transformation t-codes of FRED-MD, as shown in the data transformation tables in Appendix 6. Then I check for outliers in the transformed series following the approach of [McCracken and Ng \(2016\)](#), where an outlier is identified as an observation that deviates from the sample median by more than 10 interquartile ranges. These outliers are treated as missing values and replaced with the previous non-outlier value in the same transformed series to avoid potential look-ahead bias. Finally, four lags of all variables and four autoregressive terms of US CPI inflation are considered, giving a total of 480 predictors. The target variable in the analysis is monthly inflation, measured as the log-difference of the Consumer Price Index (CPI).

$$\pi_t = \log(P_t) - \log(P_{t-1})$$

Where π_t is the inflation at month t and P_t is a given price index at time t , and in this case, CPI is the chosen price index. Figure 1 displays the time series plot of US CPI inflation over the full sample period. The shaded areas illustrate the out-of-sample window starting from January 1990 to September 2025. In particular, gray- and red-shaded segments represent periods of low and high inflation volatility, respectively. The predictive performance of every model is evaluated on the full out-of-sample window and in two distinct subsample periods. In particular, the first subsample corresponds to less volatile inflation from January 1990 to December 1999 ($\sigma = 0.16$), in contrast to the second subsample, where the inflation is more volatile ($\sigma = 0.30$). Relative to the 1990-1999 period, the inflation rate is more volatile in 2000-2025 due to economic recessions from the global financial crisis, the COVID-19 pandemic, and the Russia-Ukraine war. Table 1 provides the descriptive statistics of the target variable on the two evaluation subsamples.

4.2 Hyperparameters Tuning

The ML models employed in this study have been shown theoretically and empirically to have outstanding performance in terms of inflation forecasting due to their inherent ability to capture complex and nonlinear relationships and to handle large datasets; however, their estimates and forecasts are highly sensitive to the hyperparameter values that heavily influence the forecasting accuracy according to [Doojav and Narmandakh \(2025\)](#). To address this issue, I use the approach in [Huang et al. \(2025\)](#) to train models and tune hyperparameters for ML methods.

Table 2 below presents the search space for ML model hyperparameters. Specifically, for all shrinkage methods, the search grids of the penalty term λ are manually set, and they are relatively different from the literature to ensure the suitability for the large training window, and the numerical convergence issue when training models in a rolling window scheme, where the algorithms have to fit thousands of models in each window per forecasting horizon. Therefore, the search space of Ridge ranges from 0.001 to 10000, while Elastic Net, LASSO, and adaptive LASSO share the same penalty grid of log-spaced values. On the other hand, the tree-based models rely on the suggested hyperparameter search space in [Doojav and Narmandakh \(2025\)](#), with only minor variations. However, there are also other constant parameters following the idea of [Medeiros et al. \(2021\)](#), such as the α parameter for the Elastic Net penalty, which is fixed to 0.5, and the default weights of the adaptive version of LASSO. The number of trees in the forest for RF $n_estimators$

is set to 500 trees, and the same holds for the maximum number of boosting trees for XGBoost with $n_estimators = 500$. A separate default minimum number of samples to be at a leaf node $min_samples_leaf$, for the Random Forest models is kept at 5, as illustrated by [Doojav and Narmandakh \(2025\)](#). The combination of search spaces for key hyperparameters with default values is employed in this inflation forecasting study to balance computational efficiency.

Table 2: Search space for tuning the hyperparameters of ML models

Model	Search Space	Fixed settings
Ridge	$\lambda \in \{0.001, 0.01, 0.1, 1, 10, 100, 1000, 10000\}$	–
LASSO	$\lambda \in 12$ log-spaced values in $[10^{-5}, 1]$	–
adaLASSO	$\lambda \in 12$ log-spaced values in $[10^{-5}, 1]$	weights $w_i = 1/(\tilde{\beta}_{i,h} + 1/\sqrt{T})$
Elastic Net	$\lambda \in 12$ log-spaced values in $[10^{-5}, 1]$	$\alpha = 0.5$
Random Forest	max features $\in \{0.1, 0.2, 0.3, 0.5\}$	trees = 500; min. leaf = 5
XGBoost	max depth $\in \{2, 4, 6\}$; learning rate $\in \{0.1, 0.2, 0.3\}$	estimators = 500
adaLASSO-RF	selection $\lambda \in [10^{-5}, 1]$; max features $\in \{0.1, 0.2, 0.3, 0.5\}$	trees = 500; min. leaf = 5
adaLASSO-XGBoost	selection $\lambda \in [10^{-5}, 1]$; max depth $\in \{2, 4, 6\}$; learning rate $\in \{0.1, 0.2, 0.3\}$	estimators = 500

A rolling-window procedure is employed for modeling and hyperparameter tuning, as in [Huang et al. \(2025\)](#). The fixed-length training window for each model estimation is the last R_h months up to time t . It makes an h -step-ahead forecast for month $t + h$, then moves the window forward by one month, $t + 1$, while holding the in-sample size constant; this repeats throughout the sample. As shown by Figure 2, there are two loops, where the outer loop is a series of rolling windows to train and test the model. Each training window, in chronological order, is split into a training subset (80%) and a validation set (20%) for tuning hyperparameters in ML models. The reason for this 80% cutoff within the inner loop is that this partition provides a reasonable proportion of data for training the model while ensuring a sufficient portion for model validation, and this approach is common in macroeconomic time series forecasting, as advocated by [Doojav and Narmandakh \(2025\)](#). I first fit each ML model to the training subset for every possible hyperparameter value in the predefined search space in Table 2 and select the optimal parameter based on the lowest validation RMSE on each validation set. Finally, the model is refit on the entire training window using the chosen hyperparameter to produce an inflation forecast. This procedure guarantees that every model is trained on past data and tested on unseen data from subsequent months to assess inflation-forecasting performance with its chosen hyperparameters. I also use the *joblib* package to import the parallel function to speed up training and tuning of computationally expensive ML models, including RF, XGBoost, adaLASSO-RF, and adaLASSO-XGBoost.

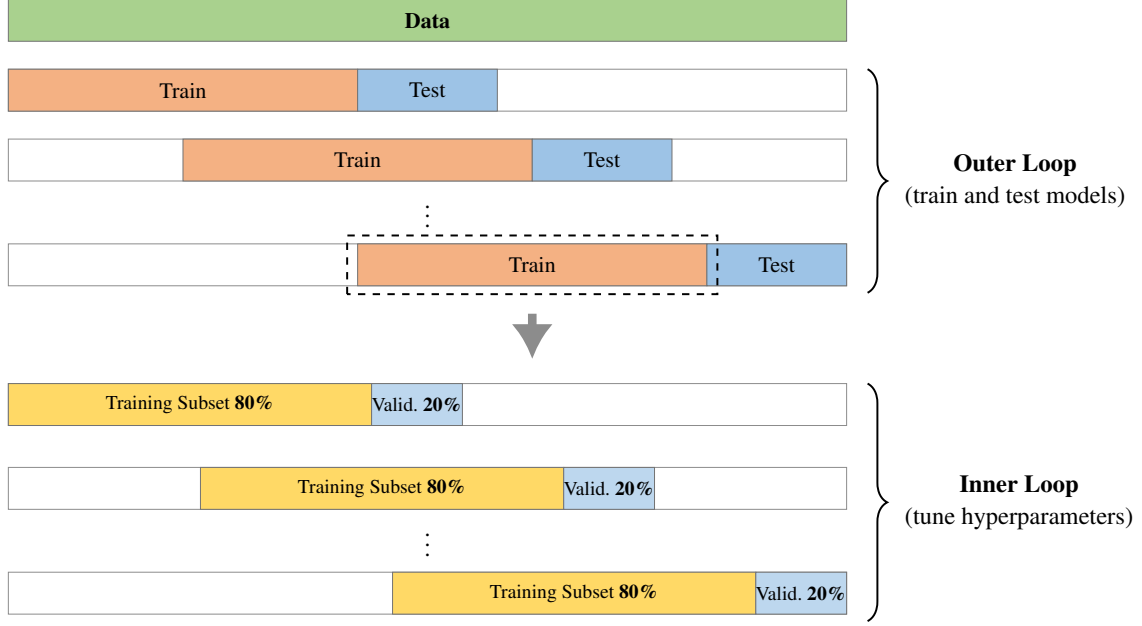


Figure 2: Rolling Window Procedure According to [Huang et al. \(2025\)](#)

5 Empirical Results

5.1 Evaluation Metrics

Following [Medeiros et al. \(2021\)](#), the out-of-sample forecasting performance of all models is compared and evaluated using the root mean squared error (RMSE), the mean absolute error (MAE), and the median absolute deviation from the median (MAD), which are formulated as follows

$$\text{RMSE}_{m,h} = \sqrt{\frac{1}{T - T_0 + 1} \sum_{t=T_0}^T \hat{u}_{t,m,h}^2},$$

$$\text{MAE}_{m,h} = \frac{1}{T - T_0 + 1} \sum_{t=T_0}^T |\hat{u}_{t,m,h}|, \quad \text{and}$$

$$\text{MAD}_{m,h} = \text{median} [|\hat{u}_{t,m,h} - \text{median}(\hat{u}_{t,m,h})|].$$

Where $\hat{u}_{t,m,h} = \hat{\pi}_{t,m,h} - \pi_t$ is the forecasting error, $\hat{\pi}_{t,m,h}$ is the inflation forecast at month t constructed by model m using available information up to $t - h$, T_0 and T represents the start and end of the out-of-sample evaluation period, and the divisor $T - T_0 + 1$ is the number of out-of-sample inflation forecasts. MAD is considered alongside RMSE and MAE because of its robustness to outliers and asymmetries. This additional evaluation statistic further confirms that some large errors do not completely determine the prediction results. In addition, the Diebold-Mariano (DM) test from [Diebold and Mariano \(2002\)](#) is used to assess whether the US CPI inflation forecasts from two distinct models are significantly different.

Table 3: Forecasting results relative to RW: Full out-of-sample (1990–2025).

Panel (a): RMSE ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	1.051	0.778	0.763	0.752	0.737
AORW	0.960	0.767	0.757	0.763	0.747
GAS	1.022	0.763	0.758	0.752	0.744
Ridge	1.000	0.784	0.754	0.754	0.725
LASSO	1.029	0.803	0.756	0.740	0.731
adaLASSO	1.031	0.773	0.745	0.751	0.707
ElasticNet	1.048	0.781	0.772	0.755	0.713
RF	0.979	0.727	0.714	0.708	0.684
XGBoost	1.073	0.788	0.755	0.725	0.786
adaLASSO-RF	1.082	0.746	0.748	0.772	0.701
adaLASSO-XGB	1.176	0.788	0.794	0.830	0.749

Panel (b): MAE ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	1.009	0.748	0.760	0.775	0.741
AORW	0.939	0.754	0.764	0.774	0.733
GAS	0.986	0.731	0.762	0.763	0.737
Ridge	0.970	0.755	0.767	0.770	0.726
LASSO	0.988	0.769	0.760	0.755	0.713
adaLASSO	0.997	0.758	0.760	0.762	0.687
ElasticNet	1.006	0.759	0.771	0.763	0.697
RF	0.954	0.721	0.721	0.712	0.667
XGBoost	1.041	0.795	0.772	0.733	0.772
adaLASSO-RF	0.995	0.732	0.765	0.790	0.694
adaLASSO-XGB	1.073	0.785	0.833	0.854	0.767

Panel (c): MAD ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	0.945	0.738	0.719	0.784	0.661
AORW	0.931	0.776	0.736	0.742	0.653
GAS	0.965	0.693	0.717	0.736	0.660
Ridge	0.988	0.756	0.765	0.811	0.721
LASSO	0.961	0.785	0.733	0.771	0.647
adaLASSO	0.974	0.793	0.736	0.718	0.642
ElasticNet	0.927	0.776	0.745	0.749	0.649
RF	0.947	0.725	0.699	0.703	0.637
XGBoost	1.038	0.849	0.749	0.730	0.720
adaLASSO-RF	0.949	0.729	0.739	0.740	0.656
adaLASSO-XGB	0.946	0.779	0.855	0.780	0.725

Table 4: Forecasting results relative to RW: Less-volatile subsample (1990–1999).

Panel (a): RMSE ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	0.898	0.876	0.800	0.910	0.888
AORW	0.785	0.800	0.776	0.826	0.729
GAS	0.846	0.853	0.831	0.942	0.867
Ridge	0.906	0.893	0.846	0.895	0.830
LASSO	0.873	0.870	0.816	0.907	0.832
adaLASSO	0.896	0.891	0.816	0.930	0.810
ElasticNet	0.892	0.885	0.848	0.898	0.820
RF	0.859	0.832	0.783	0.820	0.710
XGBoost	0.899	0.869	0.851	0.843	0.810
adaLASSO-RF	0.822	0.807	0.770	0.935	0.760
adaLASSO-XGB	0.898	0.847	0.792	0.933	0.869

Panel (b): MAE ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	0.949	0.933	0.862	1.032	1.002
AORW	0.799	0.808	0.777	0.863	0.746
GAS	0.882	0.907	0.925	1.083	0.984
Ridge	0.993	0.953	0.934	0.990	0.901
LASSO	0.935	0.926	0.910	1.003	0.852
adaLASSO	0.965	0.954	0.901	1.048	0.831
ElasticNet	0.968	0.939	0.947	1.013	0.833
RF	0.928	0.913	0.837	0.904	0.764
XGBoost	0.937	0.930	0.902	0.883	0.841
adaLASSO-RF	0.863	0.879	0.833	1.038	0.807
adaLASSO-XGB	0.945	0.922	0.847	0.979	0.934

Panel (c): MAD ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	0.784	0.630	0.703	0.867	0.636
AORW	0.695	0.706	0.727	0.889	0.659
GAS	0.749	0.684	0.668	0.826	0.593
Ridge	0.793	0.804	0.747	0.885	0.760
LASSO	0.800	0.830	0.707	0.884	0.756
adaLASSO	0.810	0.821	0.705	0.902	0.645
ElasticNet	0.844	0.790	0.701	0.869	0.723
RF	0.632	0.558	0.636	0.755	0.638
XGBoost	0.754	0.811	0.733	0.740	0.819
adaLASSO-RF	0.686	0.699	0.645	0.843	0.676
adaLASSO-XGB	0.688	0.901	0.825	0.803	0.870

Table 5: Forecasting results relative to RW: More-volatile subsample (2000–2025).

Panel (a): RMSE ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	1.070	0.769	0.759	0.739	0.720
AORW	0.981	0.764	0.755	0.758	0.749
GAS	1.043	0.755	0.751	0.735	0.730
Ridge	1.011	0.775	0.744	0.741	0.713
LASSO	1.048	0.797	0.750	0.726	0.720
adaLASSO	1.048	0.763	0.738	0.735	0.696
ElasticNet	1.068	0.771	0.765	0.742	0.702
RF	0.994	0.718	0.707	0.698	0.682
XGBoost	1.094	0.781	0.746	0.715	0.783
adaLASSO-RF	1.112	0.741	0.746	0.758	0.695
adaLASSO-XGB	1.208	0.783	0.794	0.821	0.736

Panel (b): MAE ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	1.023	0.714	0.740	0.728	0.686
AORW	0.972	0.745	0.762	0.757	0.730
GAS	1.011	0.698	0.730	0.703	0.685
Ridge	0.964	0.719	0.733	0.729	0.690
LASSO	1.001	0.741	0.730	0.709	0.684
adaLASSO	1.004	0.721	0.731	0.709	0.657
ElasticNet	1.015	0.726	0.735	0.716	0.668
RF	0.960	0.686	0.698	0.677	0.646
XGBoost	1.066	0.770	0.746	0.705	0.758
adaLASSO-RF	1.026	0.705	0.751	0.744	0.670
adaLASSO-XGB	1.103	0.760	0.830	0.831	0.732

Panel (c): MAD ratio					
Model	h=1	h=3	h=6	h=9	h=12
RW	1.000	1.000	1.000	1.000	1.000
AR	1.034	0.754	0.729	0.700	0.622
AORW	1.035	0.749	0.712	0.700	0.671
GAS	1.050	0.690	0.706	0.625	0.625
Ridge	1.020	0.673	0.755	0.712	0.680
LASSO	1.004	0.782	0.696	0.685	0.631
adaLASSO	0.982	0.751	0.689	0.622	0.564
ElasticNet	1.021	0.727	0.712	0.634	0.620
RF	0.941	0.746	0.688	0.643	0.574
XGBoost	1.033	0.832	0.704	0.667	0.672
adaLASSO-RF	0.998	0.689	0.740	0.646	0.622
adaLASSO-XGB	1.035	0.759	0.832	0.790	0.682

5.2 Interpretation

5.2.1 Full sample analysis

Table 3 reports the results of the comparison of all models in terms of the RMSE, MAE, and MAD ratio with respect to the RW benchmark for all forecasting horizons for the full out-of-sample period from 1990 to 2025. The best-performing models with the lowest ratio values are highlighted in bold. In terms of RMSE ratio, it is clear that RF achieves the highest forecasting accuracy for US CPI inflation across medium- to long-term horizons, particularly at $h = 3, 6, 9$, and 12 . However, for one-month-ahead forecasting, a majority of models are outperformed by the RW model at the short horizon $h = 1$, except for the AORW model, which has the best predictive power, with an RMSE ratio of 0.960 , followed by 0.979 of RF model. All the penalized linear regression models have almost the same forecasting performance. The hybrid specifications adaLASSO-RF and adaLASSO-XGBoost slightly worsen the forecasting accuracy with slightly higher ratio values; thus, they do not show any significant improvement in any of the performance criteria considered, compared to the standalone RF and XGBoost models, indicating that variable selection before model estimation is not useful for increasing the quality of predictions in US CPI inflation. A similar pattern emerges in the MAE ratio panel, where RF remains superior to the competitors across most horizons, and AORW performs best at $h = 1$. On the other hand, the result varies when considering the MAD ratio, as almost all models have a lower MAD ratio below 1 relative to RW, implying that the RW model delivers the worst forecasts most of the time. In particular, Elastic Net and GAS models have the most precise predictions for short- and medium-term forecasts, with Elastic Net achieving a MAD ratio of 0.927 at $h = 1$ and GAS 0.693 at $h = 3$. RF still exhibits the strongest predictive power for the remaining horizons. Another key finding across these three panels is that the performance of the RW deteriorates as the forecasting horizon increases.

5.2.2 Subsample Analysis: High-Volatility vs Low-Volatility Periods

To examine whether ML models perform well during periods of high inflation volatility, where inflation forecasts are difficult to obtain. Subsample analyses are further conducted in this section. Tables 4 and 5 present RMSE, MAE, and MAD ratios relative to the RW model during the 1990-1999 and 2000-2025 periods, respectively. The results indicated that all reported ratios of ML models are lower during turbulent periods of inflation, suggesting that most models perform worse on US CPI inflation during low-volatility inflation periods relative to the RW model. ML models are capable of maintaining high inflation-forecasting performance even under challenging, fluctuating conditions. The superiority of RF is more obvious during the 2000-2025 period, where the inflation rate is more volatile. During this period, RF achieves the lowest RMSE, MAE, and MAD across almost all forecasting horizons. In terms of the RMSE ratio, RF still confirms its dominance on medium and long horizons during high-volatility inflation periods, which exactly matches the full out-of-sample findings from the first two panels of RMSE, MAE ratio values of Table 3. The AORW has the strongest predictive performance for short- and medium-term forecasts at $h = 1$ and $h = 3$ when inflation is less volatile and at $h = 1$ when inflation is more volatile in terms of RMSE ratio. During the calm periods, adaLASSO-RF achieves the highest accuracy with an RMSE ratio of 0.770 at $h = 6$. Another point worth considering is that the time series model AORW outperforms other forecasting methods, with the smallest MAE ratio values across all hori-

zons during less-volatile period of inflation. This happens because low-volatility inflation is easier to capture by traditional benchmarks, which mostly rely on historical information to make inflation forecasts. By contrast, the superiority of non-linear ML models like RF tends to produce more precise forecasts during periods of high volatility due to their inherent ability to handle macroeconomic uncertainties arising from US inflation dynamics, as discussed by [Medeiros et al. \(2021\)](#) and [Doojav and Narmandakh \(2025\)](#). In this case, RF demonstrates the strongest forecasting performance among its competing models, achieving the lowest MAE ratios relative to RW across all horizons. Among the shrinkage models in both panels (a) and (b) during high- and low-inflation volatility periods, there is no clear evidence of a single winner because the figures for RMSE and MAE ratios are relatively similar. Moreover, other models outperform RF in terms of the MAD ratio. For example, the two regularized regression models, Ridge and adaptive LASSO, achieve MAD ratios of 0.673 for $h = 3$ and 0.622 and 0.564 for $h = 9$ and 12, respectively, during periods of fluctuating inflation. For nine-month-ahead inflation forecasting, another tree-based model, namely XGBoost, slightly outperformed RF, as indicated by a less-volatile MAD ratio of 0.740, lower than the 0.755 of RF at horizon $h = 9$. The same applies to the GAS model for twelve-month forecasts. Furthermore, the plain tree-based models outperform their hybrid counterparts across most panels in both subsample periods; therefore, this result closely mirrors the full-sample result in Table 3.

Model	Forecasting precision					
	Less Volatile (1990–1999)			More Volatile (2000–2025)		
	Aver RMSE	Aver MAE	Aver MAD	Aver RMSE	Aver MAE	Aver MAD
RW	1.000	1.000	1.000	1.000	1.000	1.000
AR(p)	0.874	0.956	0.719	0.796	0.764	0.749
AORW	0.781	0.796	0.729	0.791	0.783	0.756
GAS(1,1)	0.867	0.957	0.699	0.789	0.751	0.719
Ridge	0.872	0.952	0.795	0.784	0.755	0.752
LASSO	0.858	0.922	0.792	0.794	0.759	0.742
adaLASSO	0.866	0.935	0.771	0.781	0.750	0.702
Elastic Net	0.867	0.936	0.782	0.794	0.758	0.724
RF	0.797	0.865	0.640	0.746	0.720	0.702
XGBoost	0.853	0.896	0.773	0.808	0.795	0.764
adaLASSO-RF	0.816	0.880	0.705	0.792	0.764	0.721
adaLASSO-XGB	0.866	0.924	0.818	0.848	0.836	0.804

Table 6: Average RMSE, MAE, and MAD across all horizons relative to the RW under the less-volatile and more-volatile subsamples. The average ratio values for the best-performing models are highlighted in bold.

The forecasting results in Table 6 show the average RMSE, MAE, and MAD across all models and horizons for the less- and more-volatile subsample periods, relative to the RW model. In particular, the forecasting results align with the previous findings in Tables 3, 4, and 5, which show that ML models generally outperform the classical time-series benchmarks. In addition, the dominance and superior predictive ability of the RF model at longer forecasting horizons, especially during more volatile periods, and this result is further in line with the study of [Medeiros](#)

et al. (2021). The benchmark AORW is the best-performing model with respect to less-volatile average RMSE and MAE at short and medium horizons. A majority of the models are worse at predicting US CPI inflation during periods of low volatility. Hybrid specifications of ML models underperform relative to standalone RF and XGBoost, strongly suggesting that variable selection does not improve the accuracy of ML models for inflation forecasting.

Benchmark	Forecasting horizon				
	$h = 1$	$h = 3$	$h = 6$	$h = 9$	$h = 12$
<i>Panel (a): Squared loss</i>					
RW	-0.29	-3.59***	-2.72***	-3.10***	-3.72***
AR(p)	-3.06***	-1.87*	-1.80*	-1.86*	-2.62***
AORW	0.91	-1.48	-1.73*	-2.15**	-2.07**
GAS(1,1)	-1.86*	-1.66*	-1.89*	-2.84***	-3.71***
<i>Panel (b): Absolute loss</i>					
RW	-0.96	-5.36***	-4.35***	-4.05***	-5.25***
AR(p)	-2.80***	-1.64	-1.75*	-2.93***	-3.37***
AORW	0.64	-1.46	-1.57	-1.70*	-1.83*
GAS(1,1)	-1.63	-0.62	-2.15**	-3.01***	-3.52***

Table 7: Forecasting result: Diebold–Mariano test results over the full out-of-sample period (1990–2025). Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7 reports the DM test statistics results when comparing RF against each of the benchmark models for each forecasting horizon over the full out-of-sample period from 1990 to 2025. The defined loss differential here is the difference between the loss of RF and the loss of each benchmark. Overall, the DM test statistics are negative in nearly all cells, except at the one-month-ahead forecasting horizon for AORW, indicating that AORW is the only time-series model to outperform the RF model, but only insignificantly. In particular, RF has significantly better forecasting performance than RW at horizon $h = 3$ and onward under both loss functions. Still, the forecasting accuracies between RF and RW are statistically indistinguishable at $h = 1$, and this follows exactly the RMSE ratio values relative to RW of RF at 0.979 in Table 3, which is close to 1. RF significantly outperforms the AR benchmark, particularly at 1% significance level at $h = 1$ and $h = 12$ for both loss functions and for the square-loss differential, with significance at the 10% level for $h = 3$ to $h = 9$. The same idea holds for the case of the absolute loss differential at $h = 6$, except in the case of an insignificant value of -1.64 for $h = 3$. Turning attention to AORW, it produces forecasts that are insignificantly more accurate than RF at $h = 1$. However, RF outperforms AORW as the horizon increases, especially for $h = 9$ and 12 at the 5% significance level under the squared loss function and 10% under the absolute loss function. Finally, there is statistically strong evidence that RF beats the GAS model in inflation forecasting accuracy for long-term horizons, with 9-month and 12-month forecasts at the 1% significance level. These findings confirm that RF is statistically significantly superior to all four benchmark models at nine- and twelve-month forecasts. AORW remains the best forecasting method for short forecasting horizons, aligning with the results from the ratio and average RMSE, MAE, and MAD tables.

5.3 Variable Importance

I further conduct a variable importance analysis to identify the most informative predictors for predicting US CPI inflation. I compare the predictors chosen by ML models that show the best forecasting performance. Based on the analysis in section 5.2, I select the following four models: Random Forest, XGBoost, Ridge Regression, and adaptive LASSO. These ML methods have quite different characteristics. In particular, adaLASSO is a sparsity-inducing method that sets the coefficients of irrelevant variables to exactly zero, thereby achieving consistent variable selection, as shown in [Medeiros and Mendes \(2016\)](#). In contrast, RR is a non-sparse method that only shrinks the coefficients and never sets them to zero. In addition, RR and adaLASSO are linear models, whereas RF and XGBoost are highly nonlinear specifications. This set of ML methods closely follows the selection in [Medeiros et al. \(2021\)](#), which used adaLASSO, RF, and RR for variable selection. At the same time, XGBoost, RR, and RF are considered in the paper of [Doojav and Narmandakh \(2025\)](#), and RF and RR in the literature of [Huang et al. \(2025\)](#) for variable importance. To ensure that the obtained results of variable importance are comparable, I adopt the following strategy. For penalized linear regression models (RR and adaLASSO), I use the absolute values of the standardized coefficients for each variable on the same scale as a measure of variable importance, following [Huang et al. \(2025\)](#). For XGBoost, I use the gain method of [Chen and Guestrin \(2016\)](#), which is the fractional contribution of each variable to reducing prediction error, based on the information gain computed for every possible split at each node in the tree involving that feature, as mentioned in [Doojav and Narmandakh \(2025\)](#). I employ impurity-based feature importance analysis for each variable in the RF model, as introduced in [Liaw et al. \(2002\)](#). To measure and compare the importance of variable groups, I use the following aggregation steps, as described in [Huang et al. \(2025\)](#). First, I sum the four lags of each variable and the US CPI inflation lags to obtain the final contribution for each feature. Second, I categorize each resulting variable into the following eight variable groups, as suggested by [McCracken and Ng \(2016\)](#) and [Medeiros et al. \(2021\)](#) for the monthly macroeconomic FRED-MD data: (i) output and income; (ii) labor market; (iii) housing; (iv) consumption and orders; (v) money and credit; (vi) interest and exchange rates; (vii) prices; and (viii) stock market along with a separate autoregressive term for inflation lags. Third, I compute the within-group mean as the variable importance for each of the above-mentioned groups. The tables and figures below report the relative importance of each variable group for the RR, adaLASSO, RF, and XGBoost across forecasting horizons for the full sample and two subsamples, along with their corresponding average importance across all horizons and their ranks for US CPI inflation forecasts.

5.3.1 Interpretation

In general, each ML model has a different set of the top four most important variable groups across the full out-of-sample period and two subsample periods of high- and low-inflation volatility per forecasting horizon, despite a few overlapping variable groups, and RF appears to have the most consistent and stable variable importance among other models. It is clear that, AR term is the most important and dominant predictor group, as it consistently shows up among the top four groups in almost all panels, ranking number one in eight of the 12 table panels, implying that inflation lags are the most reliable explanatory factor and that the US CPI inflation is strongly persistent. In contrast, the price group has the least predictive power, which is barely in the top four and usually has the lowest rankings across the result panels, except in the case of low-inflation-volatility periods for adaptive LASSO from 1990 to 1999 in Figure 6b, with a third rank before the labor market. Another notable point is that the variable group importance result is similar for the two tree-based models. In particular, the top four are identical across the full sample (AR term, housing, labor market, interest and exchange rates). During the 1990-1999 period, when inflation is less volatile, the shared groups are AR, housing, money and credit. Furthermore, housing is often ranked second under the full sample and the less-volatile subsample for both RF and XGBoost. The interest rate and exchange rates group of predictors is also common in both the full out-of-sample period and the high-inflation-volatility period for both the tree-based models, as shown in Tables 10 and 11, and the bar and line charts of Figures 3 and 4. On the other hand, the dominance of money and credit occurs in both shrinkage models. Specifically, this predictor group is ranked first in the full and high-volatility panels when using Ridge Regression and ranked first in the case of adaptive LASSO during the period of high-volatility inflation, as shown in Tables 8 and 9 and Figures 5 and 6. This contrasts with tree-based models, which rarely include money and credit in the top four, whereas RF and XGBoost instead place a noticeable emphasis on AR terms and housing. Moreover, Ridge excludes AR terms from the top four predictor groups and ranks it in seventh position with an average importance of 8.9% across horizons, slightly above the price and housing groups during the turbulent periods of inflation from 2000 to 2025, as illustrated by the right line chart of Figure 5b. Another observation arises from the middle panel in the less-volatile inflation periods of the line chart in Figure 6b, where output and income peak at 67.5% as the highest figure across all panels at the long-term forecasting horizon $h = 9$ and are the top predictor group with a mean importance of 20.3% across all horizons and therefore outperform the inflation lags.

Group	Forecasting horizon					Avg	Rank
	$h = 1$	$h = 3$	$h = 6$	$h = 9$	$h = 12$		
<i>Panel (a): Full out-of-sample 1990–2025</i>							
AR (inflation lags)	5.9%	6.3%	19.5%	15.0%	11.6%	11.6%	4
Output & income	12.7%	13.0%	9.9%	8.8%	7.3%	10.4%	6
Labor market	13.8%	14.6%	11.9%	11.4%	14.1%	13.1%	2
Housing	8.1%	10.6%	5.5%	6.3%	7.0%	7.5%	9
Consumption & orders	13.8%	9.5%	9.3%	10.2%	9.8%	10.5%	5
Money & credit	13.1%	17.4%	15.2%	14.8%	16.2%	15.4%	1
Interest & exch. rates	11.5%	11.3%	8.0%	7.7%	10.3%	9.8%	7
Prices	8.7%	7.1%	6.7%	10.5%	10.3%	8.7%	8
Stock market	12.4%	10.1%	13.9%	15.2%	13.4%	13.0%	3
<i>Panel (b): High volatility 2000–2025</i>							
AR (inflation lags)	2.5%	4.6%	11.6%	12.7%	13.3%	8.9%	7
Output & income	13.0%	14.2%	11.0%	9.4%	7.8%	11.1%	5
Labor market	13.8%	14.0%	12.4%	10.8%	13.6%	12.9%	3
Housing	9.4%	11.7%	6.1%	6.6%	8.2%	8.4%	9
Consumption & orders	13.6%	9.7%	9.9%	11.1%	8.4%	10.5%	6
Money & credit	13.5%	17.4%	15.5%	15.3%	15.4%	15.4%	1
Interest & exch. rates	12.7%	13.3%	10.0%	9.6%	10.6%	11.2%	4
Prices	8.1%	6.8%	7.3%	10.0%	10.3%	8.5%	8
Stock market	13.5%	8.3%	16.1%	14.7%	12.4%	13.0%	2
<i>Panel (c): Low volatility 1990–1999</i>							
AR (inflation lags)	18.8%	14.7%	22.1%	20.7%	15.2%	18.3%	1
Output & income	9.3%	8.3%	10.1%	10.9%	9.9%	9.7%	6
Labor market	12.4%	13.7%	11.5%	10.3%	10.4%	11.7%	3
Housing	10.1%	10.8%	11.1%	4.9%	5.6%	8.5%	9
Consumption & orders	8.4%	9.0%	9.6%	11.2%	12.7%	10.2%	4
Money & credit	9.9%	13.1%	10.4%	12.9%	14.5%	12.2%	2
Interest & exch. rates	12.1%	12.2%	9.7%	8.0%	6.3%	9.7%	8
Prices	8.5%	8.6%	7.4%	11.2%	12.8%	9.7%	7
Stock market	10.6%	9.5%	8.0%	10.0%	12.5%	10.1%	5

Table 8: Variable importance for US CPI inflation based on the Ridge regression. The last two columns report the average importance across horizons and its rank. The four most important variable groups per panel are highlighted in bold.

Group	Forecasting horizon					Avg	Rank
	$h = 1$	$h = 3$	$h = 6$	$h = 9$	$h = 12$		
<i>Panel (a): Full out-of-sample 1990–2025</i>							
AR (inflation lags)	0.0%	0.0%	64.1%	19.1%	55.5%	27.7%	1
Output & income	15.3%	25.0%	2.5%	6.9%	0.0%	9.9%	5
Labor market	18.3%	19.7%	4.0%	7.8%	10.1%	12.0%	3
Housing	3.1%	5.9%	2.8%	2.6%	6.8%	4.2%	8
Consumption & orders	24.2%	17.2%	2.6%	9.5%	5.4%	11.8%	4
Money & credit	5.7%	25.5%	13.5%	26.3%	17.7%	17.7%	2
Interest & exch. rates	10.2%	4.8%	0.5%	3.9%	4.4%	4.8%	7
Prices	1.1%	1.9%	0.5%	7.8%	0.1%	2.3%	9
Stock market	22.1%	0.0%	9.6%	15.9%	0.0%	9.5%	6
<i>Panel (b): High volatility 2000–2025</i>							
AR (inflation lags)	0.0%	0.0%	37.8%	0.0%	23.1%	12.2%	2
Output & income	17.7%	24.9%	6.0%	0.0%	3.5%	10.4%	4
Labor market	15.9%	12.8%	5.3%	7.5%	13.2%	10.9%	3
Housing	2.3%	7.9%	4.0%	28.5%	8.5%	10.2%	6
Consumption & orders	8.0%	8.0%	6.3%	10.1%	2.2%	6.9%	8
Money & credit	15.2%	28.4%	17.0%	33.9%	31.6%	25.2%	1
Interest & exch. rates	10.8%	11.5%	2.1%	12.1%	12.4%	9.8%	7
Prices	3.2%	2.6%	1.6%	7.8%	4.7%	4.0%	9
Stock market	27.0%	3.9%	20.0%	0.0%	0.8%	10.3%	5
<i>Panel (c): Low volatility 1990–1999</i>							
AR (inflation lags)	51.5%	0.0%	49.1%	0.0%	0.0%	20.1%	2
Output & income	6.7%	8.3%	5.7%	67.5%	13.5%	20.3%	1
Labor market	12.3%	24.1%	8.4%	0.0%	11.6%	11.3%	4
Housing	1.1%	2.9%	9.6%	0.0%	0.8%	2.9%	9
Consumption & orders	1.7%	5.6%	11.8%	0.0%	17.4%	7.3%	8
Money & credit	4.1%	7.2%	4.9%	0.0%	23.0%	7.9%	6
Interest & exch. rates	10.4%	18.8%	7.7%	0.0%	2.3%	7.8%	7
Prices	3.1%	10.4%	1.7%	32.5%	15.7%	12.7%	3
Stock market	9.2%	22.6%	1.1%	0.0%	15.7%	9.7%	5

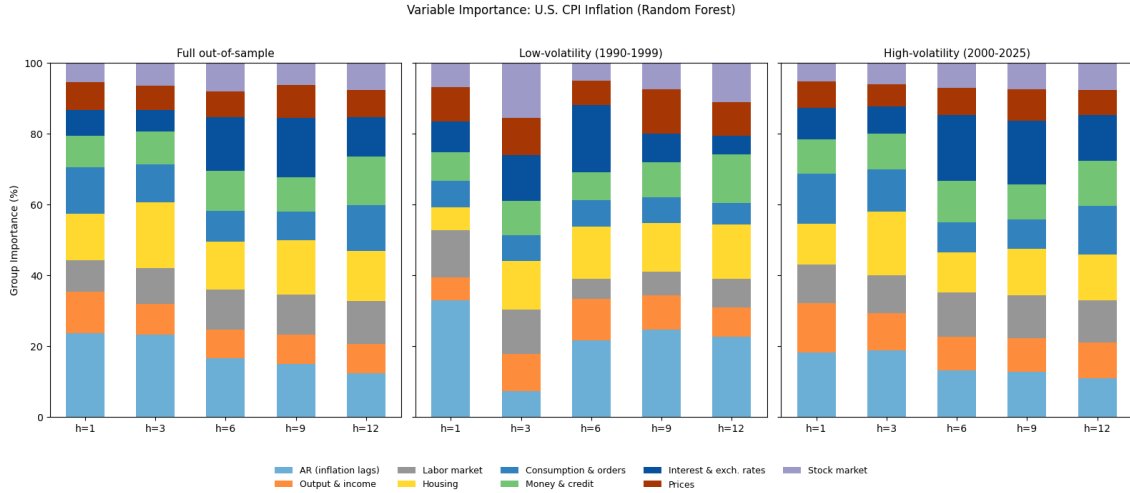
Table 9: Variable importance for US CPI inflation based on the adaptive LASSO. The last two columns report the average importance across horizons and its rank. The four most important variable groups per panel are highlighted in bold.

Group	Forecasting horizon					Avg	Rank
	$h = 1$	$h = 3$	$h = 6$	$h = 9$	$h = 12$		
<i>Panel (a): Full out-of-sample 1990–2025</i>							
AR (inflation lags)	23.5%	23.2%	16.5%	14.9%	12.2%	18.0%	1
Output & income	11.7%	8.7%	8.0%	8.2%	8.2%	9.0%	7
Labor market	8.9%	10.1%	11.3%	11.3%	12.1%	10.7%	4
Housing	13.3%	18.6%	13.7%	15.5%	14.3%	15.1%	2
Consumption & orders	13.0%	10.5%	8.5%	8.1%	13.0%	10.6%	6
Money & credit	9.0%	9.3%	11.5%	9.7%	13.6%	10.6%	5
Interest & exch. rates	7.2%	6.2%	15.0%	16.8%	11.1%	11.3%	3
Prices	7.8%	7.0%	7.3%	9.1%	7.8%	7.8%	8
Stock market	5.6%	6.5%	8.2%	6.4%	7.7%	6.9%	9
<i>Panel (b): High volatility 2000–2025</i>							
AR (inflation lags)	18.1%	18.8%	13.0%	12.7%	10.8%	14.7%	1
Output & income	13.9%	10.4%	9.6%	9.5%	10.0%	10.7%	7
Labor market	11.0%	10.6%	12.4%	12.0%	12.0%	11.6%	4
Housing	11.4%	18.0%	11.3%	13.1%	12.8%	13.3%	2
Consumption & orders	14.2%	12.1%	8.6%	8.4%	13.9%	11.4%	5
Money & credit	9.7%	10.1%	11.6%	9.7%	12.7%	10.8%	6
Interest & exch. rates	8.9%	7.7%	18.6%	18.1%	13.0%	13.2%	3
Prices	7.5%	6.3%	7.6%	9.0%	7.0%	7.5%	8
Stock market	5.3%	6.1%	7.3%	7.5%	7.8%	6.8%	9
<i>Panel (c): Low volatility 1990–1999</i>							
AR (inflation lags)	32.8%	7.2%	21.6%	24.5%	22.5%	21.7%	1
Output & income	6.6%	10.4%	11.6%	9.8%	8.3%	9.3%	6
Labor market	13.3%	12.6%	5.8%	6.5%	8.0%	9.3%	8
Housing	6.5%	13.7%	14.6%	13.8%	15.5%	12.8%	2
Consumption & orders	7.4%	7.3%	7.4%	7.3%	6.0%	7.1%	9
Money & credit	8.1%	9.6%	8.0%	9.9%	13.8%	9.9%	4
Interest & exch. rates	8.7%	13.0%	19.0%	8.0%	5.2%	10.8%	3
Prices	9.6%	10.5%	7.0%	12.4%	9.5%	9.8%	5
Stock market	7.0%	15.6%	5.1%	7.6%	11.2%	9.3%	7

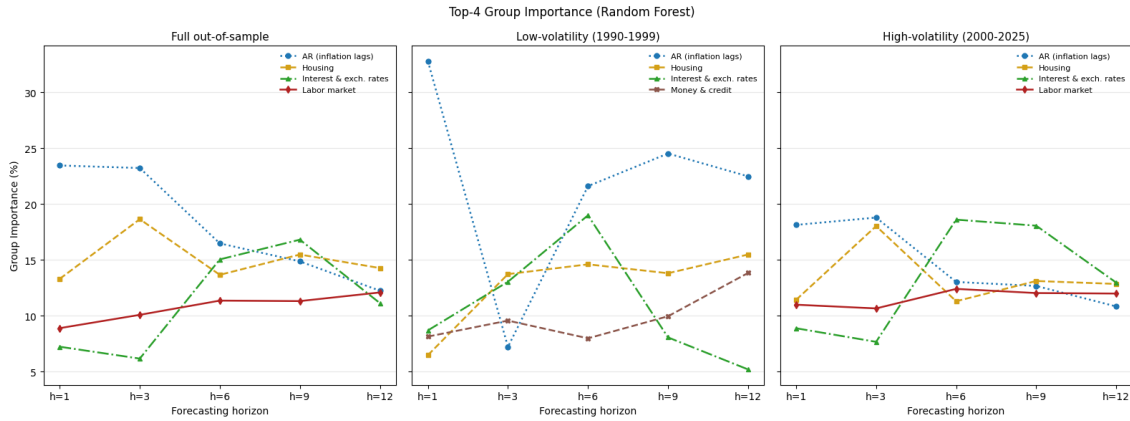
Table 10: Variable importance for US CPI inflation based on the Random Forest. The last two columns report the average importance across horizons and its rank. The four most important variable groups per panel are highlighted in bold.

Group	Forecasting horizon					Avg	Rank
	$h = 1$	$h = 3$	$h = 6$	$h = 9$	$h = 12$		
<i>Panel (a): Full out-of-sample 1990–2025</i>							
AR (inflation lags)	31.2%	17.6%	23.1%	23.1%	17.6%	22.5%	1
Output & income	9.7%	9.7%	12.0%	9.3%	7.4%	9.6%	6
Labor market	8.4%	12.0%	12.2%	8.9%	14.5%	11.2%	3
Housing	9.8%	17.4%	12.6%	14.1%	14.1%	13.6%	2
Consumption & orders	11.8%	8.7%	6.3%	7.8%	12.7%	9.5%	7
Money & credit	8.6%	9.6%	10.6%	9.3%	10.6%	9.7%	5
Interest & exch. rates	9.9%	11.3%	12.5%	8.4%	8.3%	10.1%	4
Prices	5.9%	5.6%	4.7%	8.3%	6.3%	6.2%	9
Stock market	4.8%	8.0%	6.1%	10.9%	8.5%	7.6%	8
<i>Panel (b): High volatility 2000–2025</i>							
AR (inflation lags)	7.3%	12.1%	13.7%	11.7%	25.7%	14.1%	1
Output & income	15.8%	10.1%	11.8%	12.2%	9.3%	11.8%	3
Labor market	11.9%	13.8%	12.7%	9.0%	8.3%	11.1%	5
Housing	7.7%	9.3%	9.4%	15.3%	12.3%	10.8%	6
Consumption & orders	13.9%	10.7%	8.6%	10.5%	12.3%	11.2%	4
Money & credit	11.4%	13.0%	7.1%	14.4%	7.7%	10.8%	7
Interest & exch. rates	12.9%	16.3%	13.6%	13.6%	10.5%	13.4%	2
Prices	8.4%	6.4%	5.6%	9.8%	4.0%	6.8%	9
Stock market	10.6%	8.2%	17.5%	3.4%	9.9%	9.9%	8
<i>Panel (c): Low volatility 1990–1999</i>							
AR (inflation lags)	24.4%	12.9%	28.9%	22.3%	0.0%	17.7%	1
Output & income	7.1%	8.4%	9.1%	9.5%	12.0%	9.2%	6
Labor market	9.8%	11.5%	7.0%	7.4%	9.3%	9.0%	7
Housing	8.8%	13.3%	15.8%	24.8%	23.4%	17.2%	2
Consumption & orders	16.9%	12.5%	13.6%	7.6%	8.8%	11.9%	4
Money & credit	7.9%	13.5%	8.0%	10.0%	23.2%	12.5%	3
Interest & exch. rates	5.0%	11.4%	10.2%	6.1%	7.7%	8.1%	8
Prices	10.2%	12.1%	5.6%	7.4%	15.6%	10.2%	5
Stock market	9.9%	4.3%	1.8%	4.9%	0.0%	4.2%	9

Table 11: Variable importance for US CPI inflation based on XGBoost. The last two columns report the average importance across horizons and its rank. The four most important variable groups per panel are highlighted in bold.

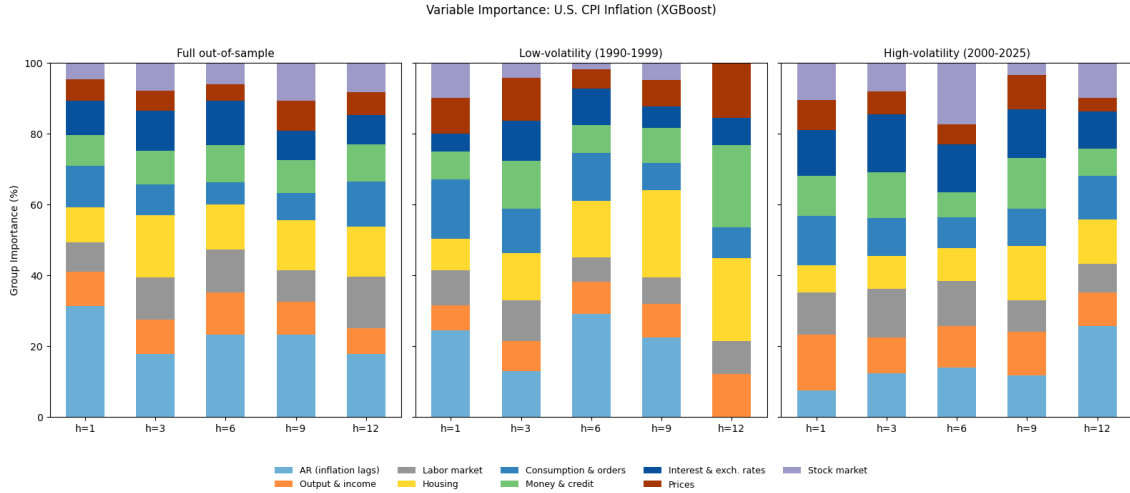


(a) The histogram for each forecasting horizon, the importance of each variable group for the RF model under the full sample and two subsample periods

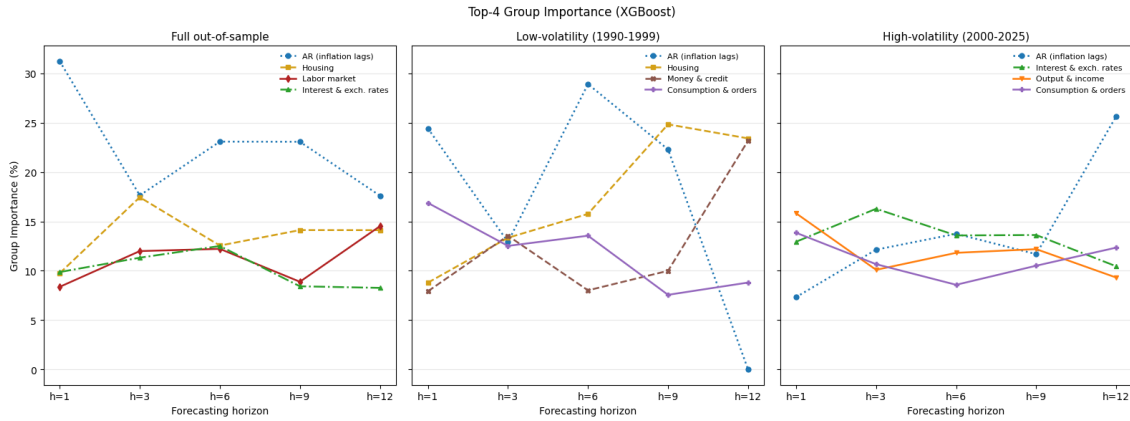


(b) The line chart for each forecasting horizon, the importance of the four most important groups for the RF model under the full sample and two subsample periods

Figure 3: Variable importance for US CPI inflation based on the Random Forest model across the full out-of-sample period and the low- and high-volatility subsample periods

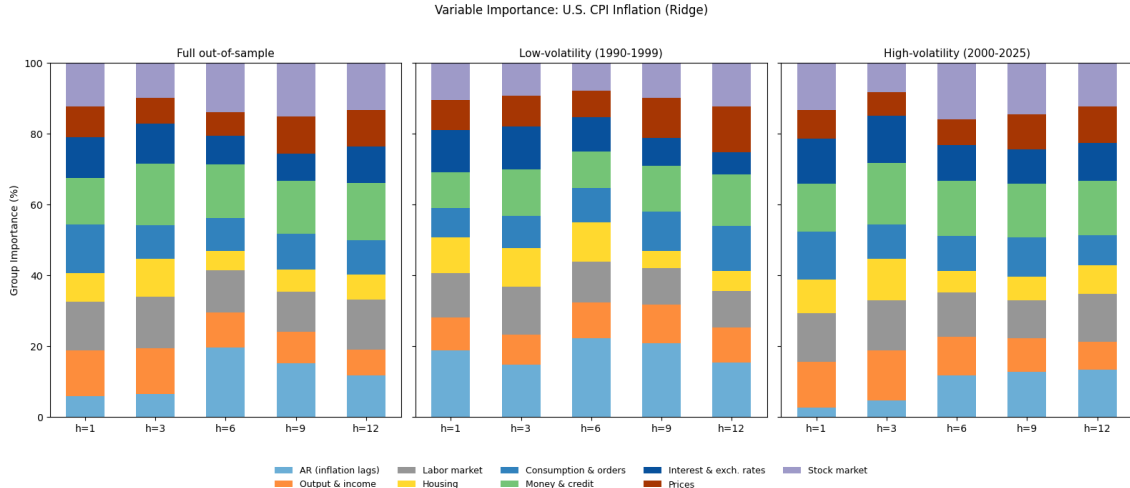


(a) The histogram for each forecasting horizon, the importance of each variable group for XGBoost under the full sample and two subsample periods

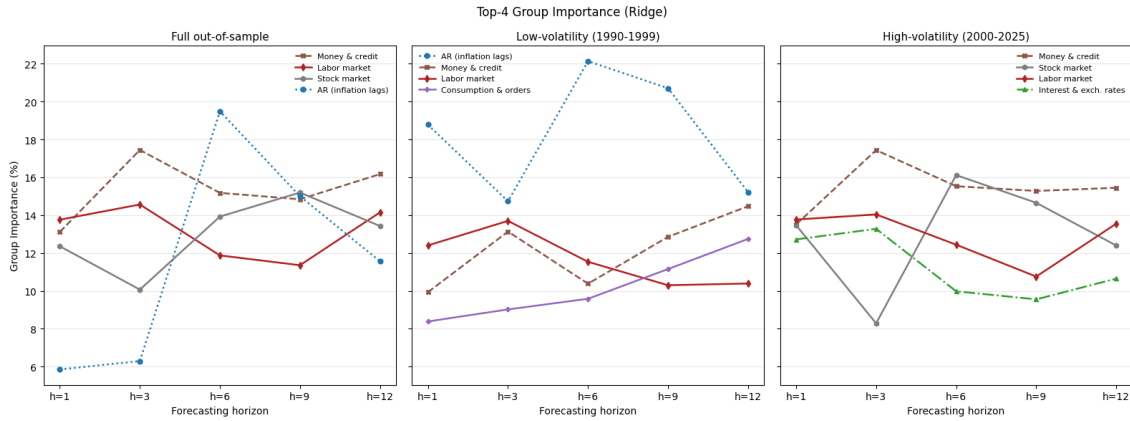


(b) The line chart for each forecasting horizon, the importance of the four most important groups for XGBoost under the full sample and two subsample periods

Figure 4: Variable importance for US CPI inflation based on the XGBoost across the full out-of-sample period and the low- and high-volatility subsample periods

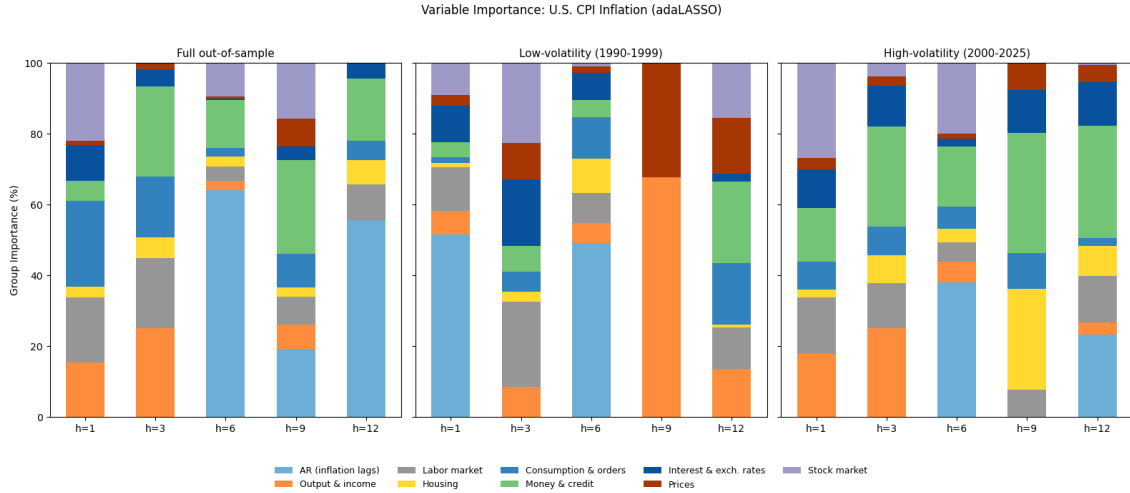


(a) The histogram for each forecasting horizon, the importance of each variable group for Ridge under the full sample and two subsample periods

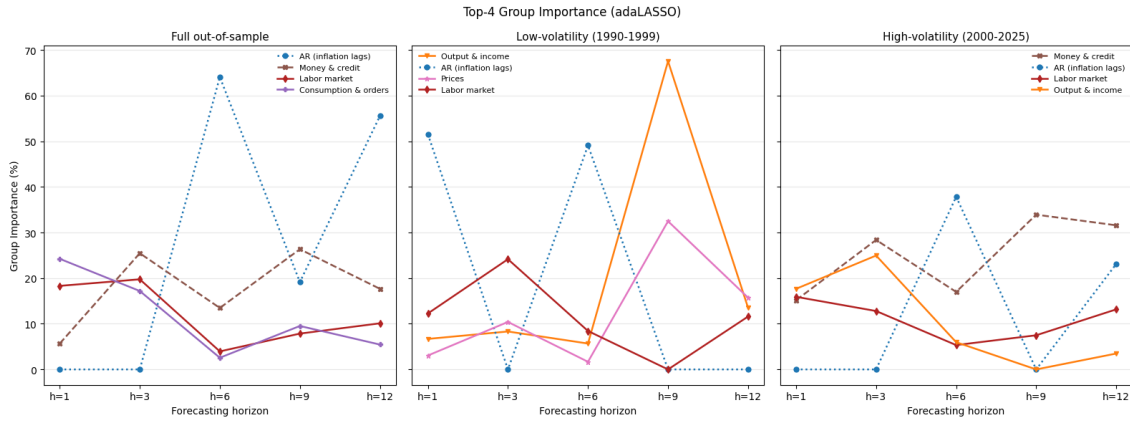


(b) The line chart for each forecasting horizon, the importance of the four most important groups for Ridge under the full sample and two subsample periods

Figure 5: Variable importance for US CPI inflation based on the Ridge Regression across the full out-of-sample period and the low- and high-volatility subsample periods



(a) The histogram for each forecasting horizon, the importance of each variable group for adaLASSO under the full sample and two subsample periods



(b) The line chart for each sample forecasting horizon, the importance of the four most important groups for adaLASSO under the full sample and two subsample periods

Figure 6: Variable importance for US CPI inflation based on the adaptive LASSO across the full out-of-sample period and the low- and high-volatility subsample periods

6 Conclusion

This thesis demonstrates the benefits of Machine Learning (ML) models in a data-rich environment for improving inflation forecasting accuracy and identifying key variables for predicting US CPI inflation. From the empirical analysis, I have the following important results. First, ML models, including Random Forest, XGBoost, Ridge Regression, and adaptive LASSO, when coupled with a large monthly macroeconomic dataset, outperform traditional benchmark time series models, particularly on medium- to long-term forecasting horizons during high-volatility inflation periods, when inflation forecasting is more difficult. The tree-based model with superior predictive performance is the Random Forest, as it has the lowest prediction errors across horizons. This can be explained by the natural variable selection mechanism and its inherent ability to capture the non-linear interactions between predictor variables and target inflation in the data. However, AORW is the special case in which the benchmark model performs better than the ML models on short- to medium-horizon forecasts, especially during less-volatile periods of inflation. This finding holds even when the Diebold-Mariano (DM) statistical test is used to evaluate the inflation-forecasting accuracy of Random Forest against each benchmark. Second, the results of the group variable importance show that the set of the most important variables selected by RF is stable across horizons in both the full out-of-sample and subsample analyses. AR terms for the inflation lags are the best predictors. Apart from that, variables from the following groups: housing, labor market, interest and exchange rates, money and credit, are also the most important predictors contributing to US CPI inflation forecasts. As a result, these variables are identified as key drivers of US inflation dynamics. Finally, the model comparison results demonstrate that the plain tree-based models RF and XGBoost produce more accurate forecasts than their hybrid ML counterparts, which have higher error rates, such as adaLASSO-RF and adaLASSO-XGBoost. This evidence implies that applying a feature selection technique before model estimation does not improve ML forecasting performance in inflation prediction. Taken together, these results provide valuable insights for ML modeling and forecasting inflation in a high-dimensional macroeconomic setting, supporting policymakers and the government in controlling inflation rates to stabilize the economy and ensure sustainable growth. For future research, it is planned to consider the forecast combination methods, such as the simple average, the trimmed average, and the median, which provides a balance of the strengths of considered models and attenuate the impact of extreme values as proposed by [Bates and Granger \(1969\)](#) to assess whether the approach of combining inflation forecasts across multiple models improves the predictive accuracy compared to a single model. It will also be better to extend the ML model set with more advanced nonlinear ML models from neural networks, including Multi-Layer Perceptron (MLP) by [Bishop \(1995\)](#), recurrent neural network (RNN) by [Elman \(1990\)](#), which have a network structure to deal with high-dimensional datasets and handle the nonlinear relationships. Furthermore, the hybrid ML specifications would use more shrinkage methods, LASSO, and Elastic Net to preselect the features, besides adaptive LASSO for tree-based and neural network models, to further verify the usefulness of feature selection in the inflation forecasting exercise.

Appendix

According to [McCracken and Ng \(2016\)](#) and [Medeiros et al. \(2021\)](#), the column t-code denotes the following stationary data transformation for a series x : (1) no transformation; (2) Δx_t ; (3) $\Delta^2 x_t$; (4) $\log(x_t)$; (5) $\Delta \log(x_t)$; (6) $\Delta^2 \log(x_t)$; (7) $\Delta(x_t/x_{t-1} - 1)$.

Group 1: Output and income						
	id	tcode	fred	description	gsi	gsi:description
1	1	5	RPI	Real Personal Income	M_14386177	PI
2	2	5	W875RX1	Real personal income ex transfer receipts	M_145256755	PI less transfers
3	6	5	INDPRO	IP Index	M_116460980	IP: total
4	7	5	IPFPNSS	IP: Final Products and Nonindustrial Supplies	M_116460981	IP: products
5	8	5	IPFINAL	IP: Final Products (Market Group)	M_116461268	IP: final prod
6	9	5	IPCONGD	IP: Consumer Goods	M_116460982	IP: cons gds
7	10	5	IPDCONGD	IP: Durable Consumer Goods	M_116460983	IP: cons dble
8	11	5	IPNCONGD	IP: Nondurable Consumer Goods	M_116460988	IP: cons nondble
9	12	5	IPBUSEQ	IP: Business Equipment	M_116460995	IP: bus eqpt
10	13	5	IPMAT	IP: Materials	M_116461002	IP: matls
11	14	5	IPDMAT	IP: Durable Materials	M_116461004	IP: dble matls
12	15	5	IPNMAT	IP: Nondurable Materials	M_116461008	IP: nondble matls
13	16	5	IPMANSICS	IP: Manufacturing (SIC)	M_116461013	IP: mfg
14	17	5	IPB51222s	IP: Residential Utilities	M_116461276	IP: res util
15	18	5	IPFUELS	IP: Fuels	M_116461275	IP: fuels
16	19	1	NAPMPI	ISM Manufacturing: Production Index	M_110157212	NAPM prodn
17	20	2	CUMFNS	Capacity Utilization: Manufacturing	M_116461602	Cap uti

Group 2: Labor market						
	id	tcode	fred	description	gsi	gsi:description
1	21*	2	HWI	Help-Wanted Index for United States		Help wanted indx
2	22*	2	HWIURATIO	Ratio of Help Wanted/No. Unemployed	M_110156531	Help wanted/unemp
3	23	5	CLF16OV	Civilian Labor Force	M_110156467	Emp CPS total
4	24	5	CE16OV	Civilian Employment	M_110156498	Emp CPS nonag
5	25	5	UNRATE	Civilian Unemployment Rate	M_110156541	U: all
6	26	2	UEMPMEAN	Average Duration of Unemployment (Weeks)	M_110156528	U: mean duration
7	27	5	UEMPLT5	Civilians Unemployed - Less Than 5 Weeks	M_110156527	U < 5 wks
8	28	5	UEMP5TO14	Civilians Unemployed for 5-14 Weeks	M_110156523	U 5-14 wks
9	29	5	UEMP15OV	Civilians Unemployed - 15 Weeks & Over	M_110156524	U 15+ wks
10	30	5	UEMP15T26	Civilians Unemployed for 15-26 Weeks	M_110156525	U 15-26 wks
11	31	5	UEMP27OV	Civilians Unemployed for 27 Weeks and Over	M_110156526	U 27+ wks
12	32*	5	CLAIMSx	Initial Claims	M_15186204	UI claims
13	33	5	PAYEMS	All Employees: Total nonfarm	M_123109146	Emp: total
14	34	5	USGOOD	All Employees: Goods-Producing Industries	M_123109172	Emp: gds prod
15	35	5	CES1021000001	All Employees: Mining and Logging: Mining	M_123109244	Emp: mining
16	36	5	USCONS	All Employees: Construction	M_123109331	Emp: const
17	37	5	MANEMP	All Employees: Manufacturing	M_123109542	Emp: mfg
18	38	5	DMANEMP	All Employees: Durable goods	M_123109573	Emp: dble gds
19	39	5	NDMANEMP	All Employees: Nondurable goods	M_123110741	Emp: nondbles
20	40	5	SRVPRD	All Employees: Service-Providing Industries	M_123109193	Emp: services
21	41	5	USTPU	All Employees: Trade, Transportation & Utilities	M_123111543	Emp: TTU
22	42	5	USWTRADE	All Employees: Wholesale Trade	M_123111563	Emp: wholesale
23	43	5	USTRADE	All Employees: Retail Trade	M_123111867	Emp: retail
24	44	5	USFIRE	All Employees: Financial Activities	M_123112777	Emp: FIRE
25	45	5	USGOVT	All Employees: Government	M_123114411	Emp: Govt
26	46	1	CES0600000007	Avg Weekly Hours : Goods-Producing	M_140687274	Avg hrs
27	47	2	AWOTMAN	Avg Weekly Overtime Hours : Manufacturing	M_123109554	Overtime: mfg
28	48	1	AWHMAN	Avg Weekly Hours : Manufacturing	M_14386098	Avg hrs: mfg
29	49	1	NAPMEI	ISM Manufacturing: Employment Index	M_110157206	NAPM empl
30	127	6	CES0600000008	Avg Hourly Earnings : Goods-Producing	M_123109182	AHE: goods
31	128	6	CES2000000008	Avg Hourly Earnings : Construction	M_123109341	AHE: const
32	129	6	CES3000000008	Avg Hourly Earnings : Manufacturing	M_123109552	AHE: mfg

Group 3: Housing						
	id	tcode	fred	description	gsi	gsi:description
1	50	4	HOUST	Housing Starts: Total New Privately Owned	M_110155536	Starts: nonfarm
2	51	4	HOUSTNE	Housing Starts, Northeast	M_110155538	Starts: NE
3	52	4	HOUSTMW	Housing Starts, Midwest	M_110155537	Starts: MW
4	53	4	HOUSTS	Housing Starts, South	M_110155543	Starts: South
5	54	4	HOUSTW	Housing Starts, West	M_110155544	Starts: West
6	55	4	PERMIT	New Private Housing Permits (SAAR)	M_110155532	BP: total
7	56	4	PERMITNE	New Private Housing Permits, Northeast (SAAR)	M_110155531	BP: NE
8	57	4	PERMITMW	New Private Housing Permits, Midwest (SAAR)	M_110155530	BP: MW
9	58	4	PERMITS	New Private Housing Permits, South (SAAR)	M_110155533	BP: South
10	59	4	PERMITW	New Private Housing Permits, West (SAAR)	M_110155534	BP: West

Group 4: Consumption, orders, and inventories						
	id	tcode	fred	description	gsi	gsi:description
1	3	5	DPCERA3M086SBEA	Real personal consumption expenditures	M_123008274	Real Consumption
2	4*	5	CMRMTSPLx	Real Manu. and Trade Industries Sales	M_110156998	M&T sales
3	5*	5	RETAILx	Retail and Food Services Sales	M_130439509	Retail sales
4	60	1	NAPM	ISM : PMI Composite Index	M_110157208	PMI
5	61	1	NAPMNOI	ISM : New Orders Index	M_110157210	NAPM new ordrs
6	62	1	NAPMSDI	ISM : Supplier Deliveries Index	M_110157205	NAPM vendor del
7	63	1	NAPMII	ISM : Inventories Index	M_110157211	NAPM Invent
8	64	5	ACOGNO	New Orders for Consumer Goods	M_14385863	Orders: cons gds
9	65*	5	AMDMNOx	New Orders for Durable Goods	M_14386110	Orders: dble gds
10	66*	5	ANDENOx	New Orders for Nondefense Capital Goods	M_178554409	Orders: cap gds
11	67*	5	AMDMUOx	Unfilled Orders for Durable Goods	M_14385946	Unf orders: dble
12	68*	5	BUSINVx	Total Business Inventories	M_15192014	M&T invent
13	69*	2	ISRATIOx	Total Business: Inventories to Sales Ratio	M_15191529	M&T invent/sales
14	130*	2	UMCSENTx	Consumer Sentiment Index	hhsntn	Consumer expect

Group 5: Money and credit						
	id	tcode	fred	description	gsi	gsi:description
1	70	6	M1SL	M1 Money Stock	M_110154984	M1
2	71	6	M2SL	M2 Money Stock	M_110154985	M2
3	72	5	M2REAL	Real M2 Money Stock	M_110154985	M2 (real)
4	73	6	AMBSL	St. Louis Adjusted Monetary Base	M_110154995	MB
5	74	6	TOTRESNS	Total Reserves of Depository Institutions	M_110155011	Reserves tot
6	75	7	NONBORRES	Reserves Of Depository Institutions	M_110155009	Reserves nonbor
7	76	6	BUSLOANS	Commercial and Industrial Loans	BUSLOANS	C&I loan plus
8	77	6	REALLN	Real Estate Loans at All Commercial Banks	BUSLOANS	DC&I loans
9	78	6	NONREVSL	Total Nonrevolving Credit	M_110154564	Cons credit
10	79*	2	CONSPI	Nonrevolving consumer credit to Personal Income	M_110154569	Inst cred/PI
11	131	6	MZMSL	MZM Money Stock	N.A.	N.A.
12	132	6	DTCOLNVHFN	Consumer Motor Vehicle Loans Outstanding	N.A.	N.A.
13	133	6	DTCTHFN	Total Consumer Loans and Leases Outstanding	N.A.	N.A.
14	134	6	INVEST	Securities in Bank Credit at All Commercial Banks	N.A.	N.A.

Group 6: Interest and exchange rates						
	id	tcode	fred	description	gsi	gsi:description
1	84	2	FEDFUNDS	Effective Federal Funds Rate	M_110155157	Fed Funds
2	85*	2	CP3Mx	3-Month AA Financial Commercial Paper Rate	CPF3M	Comm paper
3	86	2	TB3MS	3-Month Treasury Bill	M_110155165	3 mo T-bill
4	87	2	TB6MS	6-Month Treasury Bill	M_110155166	6 mo T-bill
5	88	2	GS1	1-Year Treasury Rate	M_110155168	1 yr T-bond
6	89	2	GS5	5-Year Treasury Rate	M_110155174	5 yr T-bond
7	90	2	GS10	10-Year Treasury Rate	M_110155169	10 yr T-bond
8	91	2	AAA	Moody's Seasoned Aaa Corporate Bond Yield		Aaa bond
9	92	2	BAA	Moody's Seasoned Baa Corporate Bond Yield		Baa bond
10	93*	1	COMPAPFFx	3-Month Commercial Paper Minus FEDFUNDS		CP-FF spread
11	94	1	TB3SMFFM	3-Month Treasury C Minus FEDFUNDS		3 mo-FF spread
12	95	1	TB6SMFFM	6-Month Treasury C Minus FEDFUNDS		6 mo-FF spread
13	96	1	T1YFFM	1-Year Treasury C Minus FEDFUNDS		1 yr-FF spread
14	97	1	T5YFFM	5-Year Treasury C Minus FEDFUNDS		5 yr-FF spread
15	98	1	T10YFFM	10-Year Treasury C Minus FEDFUNDS		10 yr-FF spread
16	99	1	AAAFFM	Moody's Aaa Corporate Bond Minus FEDFUNDS		Aaa-FF spread
17	100	1	BAAFFM	Moody's Baa Corporate Bond Minus FEDFUNDS		Baa-FF spread
18	101	5	TWEXMMTH	Trade Weighted U.S. Dollar Index: Major Currencies		Ex rate: avg
19	102*	5	EXSZUSx	Switzerland / U.S. Foreign Exchange Rate	M_110154768	Ex rate: Switz
20	103*	5	EXJPUSx	Japan / U.S. Foreign Exchange Rate	M_110154755	Ex rate: Japan
21	104*	5	EXUSUKx	U.S. / U.K. Foreign Exchange Rate	M_110154772	Ex rate: UK
22	105*	5	EXCAUSx	Canada / U.S. Foreign Exchange Rate	M_110154744	Ex rate: Canada

Group 7: Prices						
id	tcode	fred	description		gsi	gsi:description
1	106	5	WPSFD49207	PPI: Finished Goods	M110157517	PPI: fin gds
2	107	5	WPSFD49502	PPI: Finished Consumer Goods	M110157508	PPI: cons gds
3	108	5	WPSID61	PPI: Intermediate Materials	M_110157527	PPI: int matls
4	109	5	WPSID62	PPI: Crude Materials	M_110157500	PPI: crude matls
5	110*	5	OILPRICEx	Crude Oil, spliced WTI and Cushing	M_110157273	Spot market price
6	111	5	PPICMM	PPI: Metals and metal products	M_110157335	PPI: nonferrous
7	112	1	NAPMPRI	ISM Manufacturing: Prices Index	M_110157204	NAPM com price
8	113	5	CPIAUCSL	CPI : All Items	M_110157323	CPI-U: all
9	114	5	CPIAPPSL	CPI : Apparel	M_110157299	CPI-U: apparel
10	115	5	CPITRNSL	CPI : Transportation	M_110157302	CPI-U: transp
11	116	5	CPIMEDSL	CPI : Medical Care	M_110157304	CPI-U: medical
12	117	5	CUSR0000SAC	CPI : Commodities	M_110157314	CPI-U: comm.
13	118	5	CUUR0000SAD	CPI : Durables	M_110157315	CPI-U: dbles
14	119	5	CUSR0000SAS	CPI : Services	M_110157325	CPI-U: services
15	120	5	CPIULFSL	CPI : All Items Less Food	M_110157328	CPI-U: ex food
16	121	5	CUUR0000SA0L2	CPI : All items less shelter	M_110157329	CPI-U: ex shelter
17	122	5	CUSR0000SA0L5	CPI : All items less medical care	M_110157330	CPI-U: ex med
18	123	5	PCEPI	Personal Cons. Expend.: Chain Index	gmdc	PCE defl
19	124	5	DDURRG3M086SBEA	Personal Cons. Exp: Durable goods	gmdcd	PCE defl: dlbes
20	125	5	DNDGRG3M086SBEA	Personal Cons. Exp: Nondurable goods	gmdcn	PCE defl: nondble
21	126	5	DSERRG3M086SBEA	Personal Cons. Exp: Services	gmdcs	PCE defl: service

Group 8: Stock Market						
id	tcode	fred	description		gsi	gsi:description
1	80*	5	S&P 500	S&P's Common Stock Price Index: Composite	M_110155044	S&P 500
2	81*	5	S&P: indust	S&P's Common Stock Price Index: Industrials	M_110155047	S&P: indust
3	82*	2	S&P div yield	S&P's Composite Common Stock: Dividend Yield		S&P div yield
4	83*	5	S&P PE ratio	S&P's Composite Common Stock: Price-Earnings Ratio		S&P PE ratio
5	135*	1	VXOCLSx	VXO		

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